

A Dirichlet Pólya Inequality for Three-Dimensional Spherical Shells

Main proof manuscript

15 July 2026

Scope and status. This manuscript proves a theorem for the class of genuine three-dimensional spherical shells. It does not claim the general Pólya conjecture, literature novelty, or priority. All shell-specific analytic reductions, regional arguments, seam assignments, exact constants, and the mathematical contracts for the finite interval computations are stated in this paper. The complete executable replay sources are collected in the accompanying certification supplement. The imported background and analytic results comprise explicitly identified standard Sturm–Liouville and Bessel facts and published phase, Bessel-zero, and one-dimensional floor-sum inequalities, each quoted with its precise scope.

Abstract

For $0 < r < R$, let

$$A_{r,R} = \{x \in \mathbb{R}^3 : r < |x| < R\}.$$

We prove, with the strict Dirichlet counting convention, that

$$N_D(A_{r,R}, \Lambda) \leq L_3 |A_{r,R}| \Lambda^{3/2}, \quad L_3 = \frac{1}{6\pi^2}, \quad \Lambda \geq 0.$$

This is a computer-assisted proof. It separates variables, converts the radial spectrum into a strict multiplicity-weighted phase count, and combines analytic endpoint estimates, finite angular and radial mode inventories, and a rigorously certified final compact window using exact-rational checks and outward Arb interval certificates; the complete replay sources are provided in the accompanying certification supplement. We also give an elementary proof that no genuine spherical shell tiles $\mathbb{R}^3$ by congruent rigid-motion copies with disjoint interiors, under either exact or almost-everywhere coverage.

Contents

1	Main results and conventions	3
2	Separated spectrum, phase reduction, and the small-hole endpoint	3
2.1	The exact separated spectrum	3
2.2	Uniform phase enclosures and the strict lattice proxy	7
2.3	Multiplicity reduction to shifted one-dimensional tails	10
2.4	The low-interface estimate in the small-hole sector	12
2.5	Uniform closure of the small-hole endpoint	14
3	The thin-shell endpoint	15
3.1	The product comparison	15
3.2	The complementary-action comparison	18

4	The compact-middle argument in full detail	24
4.1	Constants and the primitive analytic envelope	24
4.2	Variational comparisons and the complete zero registry	32
4.3	The first and next angular staircases	35
4.4	The two-sided staircase	37
4.5	Combined closure of the lower components and the high staircase	38
4.6	The optical closure and the exact residual $\mathcal{D}_{20}$	43
4.7	Certified closure of the exact residual $\mathcal{D}_{20}$	44
4.8	Certificate protocol, reproducibility data, and coverage checklist	50
5	Proof of the spectral theorem	51
6	Proof that spherical shells do not tile	52
A	External inputs and verification boundary	54

1 Main results and conventions

The eigenvalues of the Dirichlet Laplacian on a bounded domain Ω are listed with multiplicity as

$$0 < \lambda_1^D(\Omega) \leq \lambda_2^D(\Omega) \leq \dots$$

Throughout we use the strict counting function

$$N_D(\Omega, \Lambda) := \#\{j : \lambda_j^D(\Omega) < \Lambda\}. \quad (1.1)$$

This convention matters at every spectral wall. We write $\mathbb{N} = \{1, 2, 3, \dots\}$ and $\mathbb{N}_0 = \{0, 1, 2, \dots\}$.

Theorem 1.1 (Spectral inequality). *For every $0 < r < R$ and every $\Lambda \geq 0$,*

$$N_D(A_{r,R}, \Lambda) \leq \frac{2}{9\pi}(R^3 - r^3)\Lambda^{3/2} = L_3|A_{r,R}|\Lambda^{3/2}, \quad L_3 = \frac{1}{6\pi^2}. \quad (1.2)$$

Theorem 1.2 (Non-tiling). *For every $0 < r < R$, no family of congruent rigid-motion copies of $A_{r,R}$ has pairwise-disjoint interiors and covers $\mathbb{R}^3$ exactly or up to a three-dimensional Lebesgue-null set. The same conclusion holds when coverage is formulated using closed shells.*

The two theorems concern literally the same complete shell family. Theorem 1.2 is logically independent of the spectral proof.

2 Separated spectrum, phase reduction, and the small-hole endpoint

Throughout this section

$$A_{\rho,1} = \{x \in \mathbb{R}^3 : \rho < |x| < 1\}, \quad 0 < \rho < 1,$$

and the Dirichlet counting function is strict:

$$N_D(A_{\rho,1}, K^2) := \#\{j : \lambda_j^D(A_{\rho,1}) < K^2\}.$$

For $x > 0$ we use the strict positive-integer count

$$[x]_< := \#\{n \in \mathbb{N} : n < x\} = \max\{0, [x] - 1\}. \quad (2.1)$$

Thus $[m]_< = m - 1$ for $m \in \mathbb{N}$. This convention is retained at every eigenfrequency, phase level, and half-integer angular wall.

2.1 The exact separated spectrum

Let $H_\rho = -\Delta_{A_{\rho,1}}^D$ be the Friedrichs operator associated with

$$\mathfrak{q}[f] = \int_{A_{\rho,1}} |\nabla f|^2 dx, \quad D(\mathfrak{q}) = H_0^1(A_{\rho,1}).$$

Choose an orthonormal spherical-harmonic basis satisfying

$$-\Delta_{\mathbb{S}^2} Y_{\ell m} = \ell(\ell + 1)Y_{\ell m}, \quad \ell \in \mathbb{N}_0, \quad -\ell \leq m \leq \ell.$$

Proposition 2.1 (Unitary decomposition and completeness). *For*

$$L_\ell = -\frac{d^2}{dr^2} + \frac{\ell(\ell+1)}{r^2}, \quad D(L_\ell) = H^2(\rho, 1) \cap H_0^1(\rho, 1),$$

one has the unitary equivalence

$$H_\rho \simeq \bigoplus_{\ell=0}^{\infty} \bigoplus_{m=-\ell}^{\ell} L_\ell. \quad (2.2)$$

The exact form domain on the right is

$$\left\{ (u_{\ell m}) : \sum_{\ell, m} \left(\|u_{\ell m}\|_2^2 + \mathfrak{a}_\ell[u_{\ell m}] \right) < \infty \right\}, \quad (2.3)$$

where

$$\mathfrak{a}_\ell[u] = \int_\rho^1 \left(|u'|^2 + \frac{\ell(\ell+1)}{r^2} |u|^2 \right) dr.$$

The exact operator domain is

$$\left\{ (u_{\ell m}) : \sum_{\ell, m} \left(\|u_{\ell m}\|_2^2 + \|L_\ell u_{\ell m}\|_2^2 \right) < \infty \right\}. \quad (2.4)$$

Each L_ℓ has positive, simple eigenvalues tending to infinity and a complete orthonormal eigenbasis. The full operator has compact resolvent.

Proof. Write $x = r\omega$, so $dx = r^2 dr d\omega$, and set

$$R_{\ell m}(r) = \int_{\mathbb{S}^2} f(r, \omega) \overline{Y_{\ell m}(\omega)} d\omega, \quad u_{\ell m}(r) = r R_{\ell m}(r).$$

Fubini, Tonelli, and Parseval on $\mathbb{S}^2$ give

$$\|f\|_{L^2(A_{\rho,1})}^2 = \sum_{\ell, m} \int_\rho^1 |R_{\ell m}(r)|^2 r^2 dr = \sum_{\ell, m} \int_\rho^1 |u_{\ell m}(r)|^2 dr.$$

Conversely, if $(u_{\ell m})$ is square summable, the finite sums

$$f_L(r, \omega) = \frac{1}{r} \sum_{\ell \leq L} \sum_{m=-\ell}^{\ell} u_{\ell m}(r) Y_{\ell m}(\omega)$$

are Cauchy in $L^2(A_{\rho,1})$, and their limit maps back to the given family. Hence the coefficient map is unitary onto the Hilbert direct sum. Because $r, 1/r \in W^{1,\infty}(\rho, 1)$, multiplication by either function is bounded on $H^1(\rho, 1)$ and preserves the two zero endpoint traces.

For a finite smooth harmonic expansion, polar coordinates and angular orthogonality yield

$$\mathfrak{q}[f] = \sum_{\ell, m} \int_\rho^1 \left(r^2 |R'_{\ell m}|^2 + \ell(\ell+1) |R_{\ell m}|^2 \right) dr.$$

If $u = rR$, then

$$r^2 |R'|^2 = \left| u' - \frac{u}{r} \right|^2 = |u'|^2 - \frac{(|u|^2)'}{r} + \frac{|u|^2}{r^2}.$$

Since

$$\frac{(|u|^2)'}{r} = \left(\frac{|u|^2}{r} \right)' + \frac{|u|^2}{r^2}$$

and $u(\rho) = u(1) = 0$, integration gives

$$\int_{\rho}^1 r^2 |R'|^2 dr = \int_{\rho}^1 |u'|^2 dr.$$

The angular term becomes $\ell(\ell+1)|u|^2/r^2$. Closing finite harmonic expansions in the form norm proves (2.3): one inclusion follows from componentwise form convergence and Fatou's lemma, while the reverse inclusion follows by first truncating the square-summable family in (ℓ, m) and then approximating its finitely many components by $C_c^\infty(\rho, 1)$. The uniqueness theorem for operators associated with closed semibounded forms gives (2.2).

For fixed ℓ , the potential $\ell(\ell+1)/r^2$ is continuous and bounded on $[\rho, 1]$. The regular one-dimensional Dirichlet form therefore has associated operator L_ℓ with the displayed $H^2 \cap H_0^1$ domain. The standard domain formula for an orthogonal operator sum gives (2.4).

The embedding $H_0^1(\rho, 1) \hookrightarrow L^2(\rho, 1)$ is compact, so every block has compact resolvent. To pass to the infinite direct sum, note that $r \leq 1$ implies

$$L_\ell \geq \ell(\ell+1), \quad \|(L_\ell + 1)^{-1}\| \leq \frac{1}{1 + \ell(\ell+1)}.$$

The resolvent of the full operator is consequently the operator-norm limit of its finite- ℓ truncations, each of which is compact. Hence the full resolvent is compact.

Regular Dirichlet Sturm–Liouville theory gives a complete sequence of simple eigenvalues in every block. They are strictly positive because

$$\mathfrak{a}_\ell[u] \geq \int_{\rho}^1 |u'|^2 dr \geq \frac{\pi^2}{(1-\rho)^2} \|u\|_2^2.$$

Simplicity also follows directly: two solutions of the same second-order equation that vanish at $r = \rho$ have proportional initial derivatives and hence are proportional by uniqueness. If $u_{\ell,n}$ is a complete radial eigenbasis, then

$$\frac{u_{\ell,n}(r)}{r} Y_{\ell m}(\omega)$$

over all (ℓ, m, n) is exactly the inverse image of a complete orthonormal coordinate basis of the Hilbert direct sum. This proves global completeness. $\square$

Proposition 2.2 (Bessel determinant and strict phase enumeration). *Put $\nu = \ell + \frac{1}{2}$. The positive eigenfrequencies in the ℓ -th radial block are precisely the positive zeros of*

$$D_{\nu,\rho}(k) := J_\nu(\rho k) Y_\nu(k) - Y_\nu(\rho k) J_\nu(k). \quad (2.5)$$

Every such zero is simple. With the continuous phase branch

$$J_\nu(x) + iY_\nu(x) = M_\nu(x) e^{i\theta_\nu(x)}, \quad M_\nu(x) > 0, \quad \theta_\nu(0+) = -\frac{\pi}{2}, \quad (2.6)$$

and

$$\Psi_{\nu,\rho}(K) = \theta_\nu(K) - \theta_\nu(\rho K), \quad \gamma_{\rho,K}(\nu) = \frac{\Psi_{\nu,\rho}(K)}{\pi},$$

the exact strict count is

$$N_D(A_{\rho,1}, K^2) = \sum_{\ell=0}^{\infty} (2\ell+1) [\gamma_{\rho,K}(\ell + \tfrac{1}{2})]_{<}. \quad (2.7)$$

The sum is finite. If K is a radial eigenfrequency in one or several angular channels, the root at K is excluded in each such channel and the excluded angular multiplicities add.

Proof. For $k > 0$, the physical radial equation has the general solution

$$R(r) = r^{-1/2}(c_1 J_\nu(kr) + c_2 Y_\nu(kr)).$$

The two Dirichlet conditions admit a nonzero coefficient vector exactly when

$$\det \begin{pmatrix} J_\nu(\rho k) & Y_\nu(\rho k) \\ J_\nu(k) & Y_\nu(k) \end{pmatrix} = D_{\nu,\rho}(k) = 0.$$

The Wronskian

$$J_\nu(x)Y'_\nu(x) - J'_\nu(x)Y_\nu(x) = \frac{2}{\pi x} \quad (2.8)$$

shows that the two entries in either row cannot vanish simultaneously. Thus the determinant criterion remains exact even if the J_ν values, or the Y_ν values, happen to vanish at both endpoints.

For root simplicity define the left-normalized solution

$$s_\ell(r, k) = \frac{\pi\sqrt{\rho}}{2}\sqrt{r} [J_\nu(\rho k)Y_\nu(kr) - Y_\nu(\rho k)J_\nu(kr)]. \quad (2.9)$$

Equation (2.8) gives

$$s_\ell(\rho, k) = 0, \quad \partial_r s_\ell(\rho, k) = 1, \quad s_\ell(1, k) = \frac{\pi\sqrt{\rho}}{2} D_{\nu,\rho}(k).$$

Let $v = \partial_k s_\ell$. Differentiating $(L_\ell - k^2)s_\ell = 0$ gives

$$(L_\ell - k^2)v = 2ks_\ell, \quad v(\rho, k) = \partial_r v(\rho, k) = 0.$$

At a positive determinant root, Lagrange's identity yields

$$\partial_k s_\ell(1, k) \partial_r s_\ell(1, k) = 2k \int_\rho^1 s_\ell(r, k)^2 dr > 0. \quad (2.10)$$

Thus $\partial_k s_\ell(1, k) \neq 0$, and the zero of the determinant is simple.

At zero energy the left-normalized solution is

$$s_\ell(r, 0) = \frac{\rho^{-\ell} r^{\ell+1} - \rho^{\ell+1} r^{-\ell}}{2\ell + 1},$$

so

$$s_\ell(1, 0) = \frac{\rho^{-\ell} - \rho^{\ell+1}}{2\ell + 1} > 0.$$

Equivalently, the standard small-argument expansions give

$$\lim_{k \downarrow 0} D_{\nu,\rho}(k) = \frac{\rho^{-\nu} - \rho^\nu}{\pi\nu} > 0. \quad (2.11)$$

Hence the limiting phase level zero is not an eigenfrequency.

From (2.6),

$$J_\nu = M_\nu \cos \theta_\nu, \quad Y_\nu = M_\nu \sin \theta_\nu,$$

and direct substitution gives the sign-sensitive factorization

$$D_{\nu,\rho}(k) = M_\nu(\rho k) M_\nu(k) \sin \Psi_{\nu,\rho}(k). \quad (2.12)$$

The Wronskian also gives

$$\theta'_\nu(x) = \frac{2}{\pi x M_\nu(x)^2} > 0.$$

Nicholson's integral representation

$$M_\nu(x)^2 = \frac{8}{\pi^2} \int_0^\infty \cosh(2\nu t) K_0(2x \sinh t) dt$$

and the strict decrease of K_0 imply that $M_\nu(x)^2$ is strictly decreasing in x . Therefore

$$\Psi'_{\nu,\rho}(k) = \frac{2}{\pi k} \left(M_\nu(k)^{-2} - M_\nu(\rho k)^{-2} \right) > 0. \quad (2.13)$$

The branch condition gives $\Psi_{\nu,\rho}(0+) = 0$, while the standard large-argument Bessel expansion gives

$$\Psi_{\nu,\rho}(k) = (1 - \rho)k + O(k^{-1}) \longrightarrow \infty$$

for fixed ν and ρ . Hence the positive determinant roots are uniquely indexed by

$$\Psi_{\ell+1/2,\rho}(k_{\ell,n}) = n\pi, \quad n = 1, 2, \dots$$

Strict monotonicity gives

$$k_{\ell,n} < K \iff n\pi < \Psi_{\ell+1/2,\rho}(K).$$

The direct-sum decomposition supplies angular multiplicity $2\ell + 1$. If the same numerical eigenvalue occurs at distinct values of ℓ , the corresponding spherical-harmonic sectors are orthogonal and their multiplicities add. Finally, $\lambda_{\ell,n} \geq \ell(\ell + 1)$, so only finitely many angular channels can contribute below K^2 . These observations prove (2.7), including every strict spectral wall. $\square$

The only standard inputs in the preceding proof are completeness of the spherical harmonics, regular one-dimensional Dirichlet Sturm–Liouville theory, the Bessel equation, the Wronskian, Nicholson's representation, and the usual small- and large-argument Bessel formulas; see [5].

2.2 Uniform phase enclosures and the strict lattice proxy

For $a > 0$, define

$$G_a(z) = \begin{cases} \frac{1}{\pi} \left(\sqrt{a^2 - z^2} - z \arccos \frac{z}{a} \right), & 0 \leq z \leq a, \\ 0, & z > a. \end{cases} \quad (2.14)$$

For $0 \leq z < a$, put

$$H_a(z) = \frac{3a^2 + 2z^2}{24\pi(a^2 - z^2)^{3/2}},$$

and define

$$\mathcal{F}_a(z) = \begin{cases} \max\{G_a(z) - H_a(z), -1/4\}, & 0 \leq z < a, \\ -1/4, & z \geq a. \end{cases} \quad (2.15)$$

We use the following published Bessel-phase result, stated here with every hypothesis needed in the sequel.

Lemma 2.3 (External uniform Bessel-phase enclosure). *With the branch (2.6), for every $a > 0$ and real $z \geq 0$,*

$$\mathcal{F}_a(z) - \frac{1}{4} < \frac{\theta_z(a)}{\pi} < G_a(z) - \frac{1}{4}. \quad (2.16)$$

If $0 < \mu < K$, $0 \leq z \leq K$, and

$$\Gamma_{K,\mu}(z) := \frac{\theta_z(K) - \theta_z(\mu)}{\pi},$$

then

$$0 < \Gamma_{K,\mu}(z) < G_K(z) - \mathcal{F}_\mu(z) \leq G_K(z) + \frac{1}{4}. \quad (2.17)$$

If $0 \leq z \leq \mu$, including $z = \mu$, then also

$$G_K(z) - G_\mu(z) < \Gamma_{K,\mu}(z) < G_K(z) - G_\mu(z) + \frac{1}{4}. \quad (2.18)$$

No value of the divergent expression $H_\mu(\mu)$ is used in the last statement.

Lemma 2.3 is Theorem 5.1 and Lemma 5.2 of [3], based on the real-order phase enclosures of [2]. The upper bound in (2.17) follows immediately by subtracting the lower bound for $\theta_z(\mu)$ from the upper bound for $\theta_z(K)$ in (2.16); the correlated two-sided estimate (2.18) is the sharper part of the annulus result. Both theorems are stated for real order, so the half-integers $z = \ell + \frac{1}{2}$ require no interpolation.

Fix now $K > 0$, set $\mu = \rho K$, and define

$$A_{\rho,K}(z) = G_K(z) - G_\mu(z). \quad (2.19)$$

For $0 \leq z \leq K$, let

$$s_\mu(z) = \begin{cases} \min\{H_\mu(z), 1/4\}, & 0 \leq z < \mu, \\ 1/4, & \mu \leq z \leq K, \end{cases} \quad U_{\rho,K}(z) = A_{\rho,K}(z) + s_\mu(z). \quad (2.20)$$

The second line, rather than H_μ , is always used at $z = \mu$.

Proposition 2.4 (Strict phase-to-lattice reduction). *For $0 < \rho < 1$ and $K > 0$,*

$$\gamma_{\rho,K}(z) < U_{\rho,K}(z) \leq A_{\rho,K}(z) + \frac{1}{4}, \quad 0 \leq z \leq K. \quad (2.21)$$

Because every G_a is zero-extended, $A_{\rho,K}$ is the piecewise action $G_K - G_{\rho K}$ below the inner interface and G_K at and above it. Consequently,

$$N_D(A_{\rho,1}, K^2) \leq \sum_{\nu_\ell < K} 2\nu_\ell [U_{\rho,K}(\nu_\ell)]_< \quad (2.22)$$

$$\leq S_{\rho,K} := \sum_{\nu_\ell < K} 2\nu_\ell \left[A_{\rho,K}(\nu_\ell) + \frac{1}{4} \right], \quad \nu_\ell = \ell + \frac{1}{2}. \quad (2.23)$$

The cutoff is strict: an order $\nu_\ell = K$ is inactive. At a true phase wall the radial count is one less than the ordinary floor; at a proxy wall the ordinary floor in (2.23) is retained. Moreover,

$$\int_0^\infty 2z A_{\rho,K}(z) dz = \frac{2}{9\pi} (1 - \rho^3) K^3. \quad (2.24)$$

Proof. For $0 \leq z < \mu$, the first upper bound in (2.17) reads

$$\gamma_{\rho,K}(z) < A_{\rho,K}(z) + G_{\mu}(z) - \mathcal{F}_{\mu}(z).$$

From (2.15),

$$G_{\mu}(z) - \mathcal{F}_{\mu}(z) = \min\{H_{\mu}(z), G_{\mu}(z) + 1/4\}.$$

The upper half of (2.18) also gives $\gamma_{\rho,K}(z) < A_{\rho,K}(z) + 1/4$. Taking the better of these two strict upper bounds, and using $G_{\mu}(z) + 1/4 \geq 1/4$, gives

$$\gamma_{\rho,K}(z) < A_{\rho,K}(z) + \min\{H_{\mu}(z), 1/4\} = U_{\rho,K}(z).$$

For $\mu \leq z \leq K$, one has $G_{\mu}(z) = 0$, and (2.17) gives

$$\gamma_{\rho,K}(z) < G_K(z) + 1/4 = U_{\rho,K}(z).$$

Orders $z \geq K$ do not contribute. Indeed, (2.16) gives

$$\frac{\theta_z(K)}{\pi} < -\frac{1}{4}, \quad \frac{\theta_z(\mu)}{\pi} > -\frac{1}{2},$$

because $G_K(z) = 0$ and $\mathcal{F}_{\mu}(z) = -1/4$. Together with phase monotonicity this yields

$$0 < \gamma_{\rho,K}(z) < \frac{1}{4} < 1,$$

so there is no positive phase level below K . This includes $z = K$.

For positive $x < y$, every positive integer strictly below x is also strictly below y , so $[x]_{<} \leq [y]_{<} \leq [y]$. Applying this observation to the exact phase count (2.7), and then using $s_{\mu} \leq 1/4$, proves (2.22)–(2.23) without changing either wall convention.

Finally, scaling $z = ax$ gives

$$\int_0^a 2zG_a(z) dz = \frac{2a^3}{\pi} \left(\int_0^1 x\sqrt{1-x^2} dx - \int_0^1 x^2 \arccos x dx \right).$$

The two elementary integrals are $1/3$ and $2/9$, respectively; hence the last expression is $2a^3/(9\pi)$. Subtracting the result with $a = \mu = \rho K$ from that with $a = K$ proves (2.24). $\square$

The necessity of retaining integerization can be displayed exactly. Put

$$m_K = \max\{0, \lceil K - \tfrac{1}{2} \rceil\}, \quad r_{\mu} = \max\{0, \lceil \mu - \tfrac{1}{2} \rceil\}.$$

Thus the active orders are $\nu_0, \dots, \nu_{m_K-1}$, and precisely the first r_{μ} of them lie strictly below μ . Define

$$E_0(a) = \sum_{\nu_{\ell} < a} 2\nu_{\ell}G_a(\nu_{\ell}) - \frac{2a^3}{9\pi}, \quad E_{\text{mp}}(\rho, K) = E_0(K) - E_0(\mu),$$

$$E_{\text{ph}} = \sum_{\ell=0}^{r_{\mu}-1} (2\ell+1) \min\{H_{\mu}(\nu_{\ell}), 1/4\} + \frac{m_K^2 - r_{\mu}^2}{4}.$$

For $x > 0$, define the strict remainder

$$\langle x \rangle_{<} := x - [x]_{<} \in (0, 1],$$

so $\langle m \rangle_< = 1$ at positive integer m , and set

$$E_{\text{frac}} = \sum_{\nu_\ell < K} 2\nu_\ell \langle U_{\rho,K}(\nu_\ell) \rangle_<.$$

Expanding $U = A + s_\mu$ term by term and using (2.24) gives the exact identity

$$\sum_{\nu_\ell < K} 2\nu_\ell [U_{\rho,K}(\nu_\ell)]_< - \frac{2}{9\pi}(1 - \rho^3)K^3 = E_{\text{mp}} + E_{\text{ph}} - E_{\text{frac}}. \quad (2.25)$$

In particular, the blanket quarter-slack cost is exactly

$$\frac{1}{4} \sum_{\ell=0}^{m_K-1} (2\ell + 1) = \frac{m_K^2}{4};$$

it cannot be discarded merely because it is of lower polynomial order for fixed ρ .

2.3 Multiplicity reduction to shifted one-dimensional tails

We record the two external floor-sum inequalities used in the small-hole argument. For integers $a < b$, write

$$\mathbf{T}(h; a, b) := \frac{1}{2} \lfloor h(a) \rfloor + \sum_{j=a+1}^{b-1} \lfloor h(j) \rfloor + \frac{1}{2} \lfloor h(b) \rfloor. \quad (2.26)$$

Lemma 2.5 (External trapezoidal floor bounds). *The following statements hold.*

1. *If h is nonnegative and concave on an integer interval $[a, b]$, then*

$$\mathbf{T}(h; a, b) \leq \int_a^b h(t) dt. \quad (2.27)$$

2. *Suppose h is nonnegative, decreasing, concave, and c -Lipschitz on an integer interval $[a, b]$, where $0 < c < 1$. If an integer $p \in [a, b)$ is the last point of the first ordinary-floor shelf, so that*

$$\lfloor h(a) \rfloor = \lfloor h(p) \rfloor > \lfloor h(p+1) \rfloor,$$

then

$$\mathbf{T}(h; a, b) \leq \int_a^b h(t) dt - \frac{1-c}{2}(b-p). \quad (2.28)$$

If no such drop occurs, set $p = b$; the basic bound in item 1 gives the same displayed formula with zero correction. This last no-drop case is a consequence of the basic proposition, not an invocation of the improved theorem at $p = b$.

3. *If g is nonnegative, decreasing, convex, and $1/2$ -Lipschitz on an integer interval $[a, b]$, if $g(b) = 0$, and if g is $1/3$ -Lipschitz on $[t_*, b]$ for some $t_* \in [a, b]$, then*

$$\mathbf{T}(g + 1/4; a, b) \leq \int_a^b g(t) dt - \frac{1}{4} \lfloor g(t_*) \rfloor. \quad (2.29)$$

4. *If g is nonnegative, decreasing, convex, and $1/2$ -Lipschitz on $[0, b]$, where $b > 0$ is arbitrary and $g(b) = 0$, then*

$$\left\lfloor g(0) + \frac{1}{4} \right\rfloor + 2 \sum_{j=1}^{\lfloor b \rfloor} \left\lfloor g(j) + \frac{1}{4} \right\rfloor \leq 2 \int_0^b g(t) dt. \quad (2.30)$$

The first assertion is Proposition 3.1 of [3]; the first-shelf improvement is its Theorem 1.4; and the third assertion is its Theorem 3.4. The fourth is the translated convex shifted-floor theorem of [1]. These published one-dimensional inequalities are the only non-special-function external inputs in what follows.

For $r \in \mathbb{N}_0$, put $x_r = r + \frac{1}{2}$ and define the coarse shifted tail

$$\mathcal{T}_r = \left\lfloor A_{\rho,K}(x_r) + \frac{1}{4} \right\rfloor + 2 \sum_{j \geq 1} \left\lfloor A_{\rho,K}(x_r + j) + \frac{1}{4} \right\rfloor. \quad (2.31)$$

All but finitely many terms vanish.

Lemma 2.6 (Weighted-lattice scaffold). *One has*

$$S_{\rho,K} = \sum_{r \geq 0} \mathcal{T}_r. \quad (2.32)$$

If, for every r with $x_r < K$,

$$\mathcal{T}_r \leq 2 \int_{x_r}^K A_{\rho,K}(z) dz, \quad (2.33)$$

then

$$S_{\rho,K} \leq \frac{2}{9\pi} (1 - \rho^3) K^3. \quad (2.34)$$

Proof. For any finitely supported nonnegative integer sequence q_ℓ ,

$$\sum_{\ell \geq 0} (2\ell + 1) q_\ell = \sum_{r \geq 0} \left(q_r + 2 \sum_{\ell > r} q_\ell \right).$$

Indeed, q_ℓ appears once with coefficient one when $r = \ell$ and appears ℓ times with coefficient two when $0 \leq r < \ell$. Taking

$$q_\ell = \left\lfloor A_{\rho,K}(\ell + \tfrac{1}{2}) + \frac{1}{4} \right\rfloor$$

proves (2.32); samples at or beyond K are zero.

Assume (2.33). Tonelli's theorem for the finite nonnegative sum gives

$$S_{\rho,K} \leq 2 \int_0^K f_3(z) A_{\rho,K}(z) dz, \quad f_3(z) = \#\{r \in \mathbb{N}_0 : r + \tfrac{1}{2} \leq z\}.$$

Let $F_3(z) = \int_0^z f_3(t) dt$. If $z = m + \frac{1}{2} + s$, where $m \in \mathbb{N}_0$ and $0 \leq s < 1$, then direct summation gives

$$\frac{z^2}{2} - F_3(z) = \frac{1}{2} \left(s - \frac{1}{2} \right)^2 \geq 0; \quad (2.35)$$

for $0 \leq z < 1/2$ the same inequality is immediate because $F_3(z) = 0$. The action $A_{\rho,K}$ is absolutely continuous, decreasing, and vanishes at K . Indeed, away from the inner interface,

$$A'_{\rho,K}(z) = \begin{cases} -\frac{1}{\pi} \left(\arccos \frac{z}{K} - \arccos \frac{z}{\mu} \right) < 0, & 0 < z < \mu, \\ -\frac{1}{\pi} \arccos \frac{z}{K} < 0, & \mu < z < K, \end{cases}$$

and the two formulas have the same limiting value at $z = \mu$. Integration by parts and (2.35) therefore give

$$2 \int_0^K f_3 A_{\rho,K} = -2 \int_0^K F_3 A'_{\rho,K} \leq - \int_0^K z^2 A'_{\rho,K}(z) dz = \int_0^K 2z A_{\rho,K}(z) dz.$$

Now apply (2.24). □

Lemma 2.7 (Tails at or above the inner interface). *If $x_r \geq \mu = \rho K$ and $x_r < K$, then*

$$\mathcal{T}_r \leq 2 \int_{x_r}^K A_{\rho,K}(z) dz.$$

Equality $x_r = \mu$ is included.

Proof. On $z \geq x_r \geq \mu$, one has $G_\mu(z) = 0$, including at $z = \mu$, so $A_{\rho,K}(z) = G_K(z)$. Set $b = K - x_r > 0$ and $g(t) = G_K(x_r + t)$ on $[0, b]$. Direct differentiation of (2.14) gives

$$G'_K(z) = -\frac{1}{\pi} \arccos \frac{z}{K} \in [-1/2, 0], \quad G''_K(z) = \frac{1}{\pi \sqrt{K^2 - z^2}} > 0.$$

Thus g is nonnegative, decreasing, convex, and $1/2$ -Lipschitz, with $g(b) = G_K(K) = 0$. Terms of (2.31) with $j > b$ vanish; if b is an integer, the terminal term is $\lfloor G_K(K) + 1/4 \rfloor = 0$. Therefore (2.30) is exactly the desired estimate. $\square$

2.4 The low-interface estimate in the small-hole sector

Define

$$\omega_0 := G_1(1/2) = \frac{\sqrt{3}}{2\pi} - \frac{1}{6}, \quad C_* := \frac{1}{2} + \frac{8\sqrt{2}}{15}. \quad (2.36)$$

Notice that $0 < \omega_0 < 1/2$. The lower inequality is equivalent to $\pi < 3\sqrt{3}$, which follows from $\pi < 4 < 3\sqrt{3}$; the upper inequality also follows directly from $0 < G_1(1/2) < G_1(0) = 1/\pi < 1/2$.

Lemma 2.8 (Low-interface shifted tails). *Assume*

$$0 < \rho < \omega_0, \quad K(\omega_0 - \rho) \geq C_*. \quad (2.37)$$

If $x_r = r + \frac{1}{2} < \mu = \rho K$, then, in fact strictly,

$$\mathcal{T}_r < 2 \int_{x_r}^K A_{\rho,K}(z) dz. \quad (2.38)$$

Proof. Fix a low tail and abbreviate $x_0 = x_r$, $A = A_{\rho,K}$. Put

$$n = \lfloor \mu - x_0 \rfloor, \quad q = x_0 + n, \quad B = \lceil K - x_0 \rceil. \quad (2.39)$$

Then

$$q \leq \mu < q + 1, \quad n < B, \quad G_K(x_0 + B) = 0.$$

The point q is the largest half-integer in the tail grid that does not exceed μ ; it need not equal $\lfloor \mu \rfloor$. A low tail also forces $\mu > x_0 \geq 1/2$, hence $\mu > 1/2$.

Let

$$h_A(t) = A(x_0 + t) + \frac{1}{4}, \quad h_K(t) = G_K(x_0 + t) + \frac{1}{4}.$$

All samples at and beyond B have floor zero, so

$$\frac{\mathcal{T}_r}{2} = \mathbf{T}(h_A; 0, B).$$

If $n \geq 1$, additivity of the trapezoidal sum at the shared endpoint and $A \leq G_K$ give

$$\frac{\mathcal{T}_r}{2} \leq \mathbf{T}(h_A; 0, n) + \mathbf{T}(h_K; n, B). \quad (2.40)$$

At the split point, the two half-weights add to the original full sample; after replacing the suffix by G_K , its half-sample can only increase. If $n = 0$, there is no concave block, and instead

$$\frac{\mathcal{T}_r}{2} \leq \mathbf{T}(h_K; 0, B). \quad (2.41)$$

For $0 < z < \mu$,

$$A''(z) = \frac{1}{\pi\sqrt{K^2 - z^2}} - \frac{1}{\pi\sqrt{\mu^2 - z^2}} < 0. \quad (2.42)$$

Thus A is concave on $[x_0, q]$, also when $q = \mu$ by continuity. For $n \geq 1$, the concave bound (2.27) gives

$$\mathbf{T}(h_A; 0, n) \leq \int_0^n A(x_0 + t) dt + \frac{n}{4}. \quad (2.43)$$

For the suffix set $g(t) = G_K(x_0 + t)$, using the zero extension after K . It is nonnegative, decreasing, convex, globally $1/2$ -Lipschitz, and $g(B) = 0$. Because $q \leq \mu = \rho K < \omega_0 K < K/2$,

$$t_* := \frac{K}{2} - x_0$$

belongs to $[n, B]$ (strictly to the right of n). On $[t_*, B]$,

$$|g'(t)| \leq \frac{1}{\pi} \arccos \frac{1}{2} = \frac{1}{3},$$

and

$$g(t_*) = G_K(K/2) = \omega_0 K.$$

Every hypothesis of (2.29) is therefore satisfied, and

$$\mathbf{T}(h_K; n, B) \leq \int_n^B G_K(x_0 + t) dt - \frac{1}{4} \lfloor \omega_0 K \rfloor. \quad (2.44)$$

The identical statement with $n = 0$ applies to (2.41).

Combining (2.43) and (2.44), or using only the latter when $n = 0$, leaves one and only one integral mismatch:

$$\delta := \int_q^\mu G_\mu(z) dz. \quad (2.45)$$

Indeed,

$$\int_{x_0}^q A(z) dz + \int_q^K G_K(z) dz = \int_{x_0}^K A(z) dz + \delta,$$

and the same identity holds for $n = 0$, where $q = x_0$. Hence

$$\frac{\mathcal{T}_r}{2} \leq \int_{x_0}^K A(z) dz + \delta - \frac{\lfloor \omega_0 K \rfloor - n}{4}. \quad (2.46)$$

The turning-point estimate of Lemma 6.2 in [3] states that

$$G_\mu(z) < \frac{(\mu - z)^{3/2}}{3\sqrt{\mu}}, \quad 0 < z < \mu. \quad (2.47)$$

If $q < \mu$, integration gives a strict first bound; if $q = \mu$, both sides below vanish. The endpoint-safe statement is

$$0 \leq \delta \leq \frac{2(\mu - q)^{5/2}}{15\sqrt{\mu}} < \frac{2\sqrt{2}}{15}, \quad (2.48)$$

because $0 \leq \mu - q < 1$ and $\mu > 1/2$. The last inequality remains strict when $q = \mu$, since then $\delta = 0$.

Finally, $x_0 \geq 1/2$ implies

$$n \leq \mu - x_0 \leq \rho K - \frac{1}{2},$$

while $\lfloor y \rfloor > y - 1$ for every real y . Therefore

$$\begin{aligned} \lfloor \omega_0 K \rfloor - n &> \omega_0 K - 1 - \left(\rho K - \frac{1}{2} \right) \\ &= K(\omega_0 - \rho) - \frac{1}{2} \\ &\geq \frac{8\sqrt{2}}{15} > 4\delta. \end{aligned}$$

Both strict inequalities remain strict when the threshold in (2.37) is attained. Substitution in (2.46) proves (2.38). $\square$

2.5 Uniform closure of the small-hole endpoint

Set

$$\rho_* := \frac{\omega_0}{1 + 2C_*} = \frac{\frac{\sqrt{3}}{2\pi} - \frac{1}{6}}{2 + \frac{16\sqrt{2}}{15}}. \quad (2.49)$$

Since $\omega_0 > 0$ and $C_* > 0$, $0 < \rho_* < \omega_0$. Its defining identity is

$$\omega_0 - \rho_* = 2C_*\rho_*, \quad \frac{\omega_0 - \rho_*}{2\rho_*} = C_*. \quad (2.50)$$

Theorem 2.9 (Complete small-hole endpoint). *For every $0 < \rho \leq \rho_*$ and every $K \geq 0$,*

$$\boxed{N_D(A_{\rho,1}, K^2) \leq \frac{2}{9\pi}(1 - \rho^3)K^3.} \quad (2.51)$$

The face $\rho = \rho_$, the interface wall $\rho K = 1/2$, and every strict spectral wall are included.*

Proof. If $K = 0$, the strict spectrum is positive, so the left side and the right side of (2.51) are both zero. Assume $K > 0$ and write $\mu = \rho K$.

First suppose $\mu \leq 1/2$. Every shifted-tail start satisfies

$$x_r = r + \frac{1}{2} \geq \frac{1}{2} \geq \mu.$$

Lemma 2.7 therefore proves (2.33) for every active tail. At the joining wall $\mu = 1/2$, the first start is exactly $x_0 = \mu$, which belongs to the inclusive high-interface lemma.

Now suppose $\mu > 1/2$. Then

$$K > \frac{1}{2\rho}. \quad (2.52)$$

For $0 < \rho \leq \rho_*$, the defining identity (2.50) gives

$$\frac{\omega_0 - \rho}{2\rho} \geq C_*; \quad (2.53)$$

equivalently, multiply $(1 + 2C_*)\rho \leq (1 + 2C_*)\rho_* = \omega_0$ by positive quantities. Since $\omega_0 - \rho > 0$, (2.52) and (2.53) imply

$$K(\omega_0 - \rho) > \frac{\omega_0 - \rho}{2\rho} \geq C_*. \quad (2.54)$$

Every start $x_r \geq \mu$ is controlled by Lemma 2.7; every start $x_r < \mu$ is controlled by Lemma 2.8, because $\rho \leq \rho_* < \omega_0$ and (2.54) supplies its threshold. Thus all active tails satisfy (2.33) in this branch as well.

Lemma 2.6 now yields

$$S_{\rho,K} \leq \frac{2}{9\pi}(1 - \rho^3)K^3,$$

and Proposition 2.4 gives $N_D(A_{\rho,1}, K^2) \leq S_{\rho,K}$, proving (2.51).

At $\rho = \rho_*$, inequality (2.53) is an equality, but the first inequality in (2.54) is strict whenever $\mu > 1/2$. Hence the endpoint is included. If instead $K(\omega_0 - \rho) = C_*$, then

$$\mu = \frac{C_*\rho}{\omega_0 - \rho} \leq \frac{1}{2}$$

by (2.53), so the point already belongs to the all-high-interface branch. No limiting argument at $\rho = 0$, no continuity of Bessel roots, and no finite numerical verification is used. $\square$

3 The thin-shell endpoint

We now prove the Pólya inequality uniformly as the shell becomes thin. In this section the counting function is strict:

$$N_D(\Omega, K^2) = \#\{j : \lambda_j(\Omega) < K^2\}.$$

We retain the global convention $[u]_< := \#\{n \in \mathbb{N} : n < u\} = \max\{0, [u] - 1\}$. Thus a spectral level lying exactly on the wall K^2 is not counted. Put

$$\varepsilon = 1 - \rho, \quad a = \varepsilon K, \quad m_\varepsilon = 1 - \varepsilon + \frac{\varepsilon^2}{3}. \quad (3.1)$$

Since $1 - \rho^3 = 3\varepsilon m_\varepsilon$, the desired right-hand side has the two equivalent forms

$$\frac{2}{9\pi}(1 - \rho^3)K^3 = \frac{1}{\varepsilon^2} \frac{2a^3}{3\pi} m_\varepsilon. \quad (3.2)$$

The proof uses the optimized phase estimate already established in Proposition 2.4. We record precisely the consequence that is used here. For $\nu_\ell = \ell + \frac{1}{2}$, let $\gamma_{\rho,K}(\nu_\ell)$ be the normalized radial Bessel phase. If $0 \leq \nu_\ell \leq K$, that proposition gives

$$\gamma_{\rho,K}(\nu_\ell) < A_{\rho,K}(\nu_\ell) + \frac{1}{4}, \quad (3.3)$$

where, after the scaling in (3.1), $A_{\rho,K}(\nu) = \mathcal{A}(\varepsilon\nu)$ and $\mathcal{A}$ is the explicit action in (3.20) below. The exact radial count in channel ℓ is

$$\#\{n \geq 1 : n < \gamma_{\rho,K}(\nu_\ell)\}. \quad (3.4)$$

In particular, (3.3) is an inequality for the strict count; it is not a replacement of that count by an ordinary floor.

3.1 The product comparison

Under the standard unitary radial transformation, the angular channel $\ell \in \mathbb{N}_0$ is the Dirichlet operator

$$L_\ell = -\frac{d^2}{dr^2} + \frac{\ell(\ell+1)}{r^2} \quad \text{in } L^2((\rho, 1), dr),$$

and it occurs with multiplicity $2\ell + 1$. The following lemma includes the direction of the min-max comparison and every strict radial and angular wall.

Lemma 3.1 (Strict product majorant). *Let $0 < \varepsilon < 1$, $a = \varepsilon K$, and define*

$$N(a) = \max \left\{ 0, \left\lceil \frac{a}{\pi} \right\rceil - 1 \right\}. \quad (3.5)$$

For $1 \leq n \leq N(a)$, put

$$b_n = \sqrt{a^2 - n^2 \pi^2}, \quad M_n = \left\lceil \sqrt{\left(\frac{b_n}{\varepsilon}\right)^2 + \frac{1}{4}} - \frac{1}{2} \right\rceil.$$

Then

$$N_D(A_{\rho,1}, K^2) \leq \mathcal{P}_\varepsilon(a) := \sum_{n=1}^{N(a)} M_n^2. \quad (3.6)$$

In particular, the strict shell count is zero for $0 \leq a \leq \pi$.

Proof. The two quadratic forms have the common domain $H_0^1(\rho, 1)$, and

$$\int_\rho^1 \left(|u'|^2 + \frac{\ell(\ell+1)}{r^2} |u|^2 \right) dr \geq \int_\rho^1 \left(|u'|^2 + \ell(\ell+1) |u|^2 \right) dr,$$

because $r \leq 1$. The min-max principle therefore gives

$$\lambda_{\ell,n} \geq \left(\frac{n\pi}{\varepsilon} \right)^2 + \ell(\ell+1), \quad n \geq 1.$$

Consequently, a shell eigenvalue in this channel can be counted below K^2 only if

$$n^2 \pi^2 + \varepsilon^2 \ell(\ell+1) < a^2. \quad (3.7)$$

The active radial integers are exactly those satisfying $n\pi < a$, whose number is (3.5). For a fixed active n , condition (3.7) is

$$\ell < \sqrt{\left(\frac{b_n}{\varepsilon}\right)^2 + \frac{1}{4}} - \frac{1}{2}.$$

Hence the allowed angular indices are $0, \dots, M_n - 1$, and their full spherical multiplicity is

$$\sum_{\ell=0}^{M_n-1} (2\ell+1) = M_n^2.$$

Summing proves (3.6).

If $a = m\pi$, the radial level $n = m$ is excluded by the strict inequality $n\pi < a$; thus $N(a) = m - 1$. At an angular wall $b_n^2/\varepsilon^2 = L(L+1)$, the displayed expression inside the ceiling is exactly L , so $M_n = L$ and the level $\ell = L$ is excluded. The jump of multiplicity $2L+1$ occurs only above the wall. Finally, no positive integer satisfies $n\pi < a$ when $0 \leq a \leq \pi$, including both endpoints, so the product majorant and hence the shell count are zero there. $\square$

Lemma 3.2 (Exact product deficit). *For $a > \pi$, let $N = N(a)$ and write uniquely*

$$\frac{a}{\pi} = N + t, \quad N \geq 1, \quad 0 < t \leq 1. \quad (3.8)$$

At $a = m\pi$, this means $(N, t) = (m - 1, 1)$. Define

$$I_0(a) = \frac{2a^3}{3\pi}, \quad S_0(a) = \sum_{n=1}^N (a^2 - n^2\pi^2), \quad D(a) = I_0(a) - S_0(a).$$

Then

$$\frac{D(a)}{\pi^2} = \frac{N^2}{2} + N \left(t^2 + \frac{1}{6} \right) + \frac{2t^3}{3}, \quad (3.9)$$

and, uniformly for $a > \pi$,

$$D(a) > \frac{2}{5}a^2. \quad (3.10)$$

Proof. The sum of the first N squares gives

$$\begin{aligned} \frac{D(a)}{\pi^2} &= \frac{2}{3}(N+t)^3 - N(N+t)^2 + \frac{N(N+1)(2N+1)}{6} \\ &= \frac{N^2}{2} + N \left(t^2 + \frac{1}{6} \right) + \frac{2t^3}{3}, \end{aligned}$$

which proves (3.9). At $a = m\pi$, its value with $(N, t) = (m - 1, 1)$ agrees with the right limit $(N, t) \rightarrow (m, 0+)$; both equal $m^2/2 + m/6$. Thus the formula owns the wall and both one-sided limits.

Subtracting $\frac{2}{5}(N+t)^2$ from (3.9) gives the exact identity

$$\begin{aligned} \frac{D(a)}{\pi^2} - \frac{2}{5} \left(\frac{a}{\pi} \right)^2 &= \frac{N^2}{10} + N \left[\left(t - \frac{2}{5} \right)^2 + \frac{1}{150} \right] \\ &\quad + \frac{2t^3}{3} - \frac{2t^2}{5}. \end{aligned} \quad (3.11)$$

For $h(t) = 2t^3/3 - 2t^2/5$, $h'(t) = 2t(t - 2/5)$, so the minimum on $0 < t \leq 1$ is $h(2/5) = -8/375$. Therefore the right-hand side of (3.11) is at least

$$\frac{N^2}{10} + \frac{N}{150} - \frac{8}{375}.$$

This expression increases for integral $N \geq 1$, and at $N = 1$ it is

$$\frac{1}{10} + \frac{1}{150} - \frac{8}{375} = \frac{32}{375} > 0.$$

This proves (3.10) without sampling. □

Proposition 3.3 (The inclusive low-action piece). *If*

$$0 < \varepsilon \leq \frac{1}{8}, \quad 0 \leq a \leq \frac{\pi}{4\varepsilon},$$

then

$$N_D(A_{\rho,1}, K^2) < \frac{1}{\varepsilon^2} \frac{2a^3}{3\pi} m_\varepsilon \quad (a > 0), \quad (3.12)$$

whereas both sides are zero at $a = 0$.

Proof. For $0 < a \leq \pi$, Lemma 3.1 gives a zero left-hand side and the right-hand side of (3.12) is positive. It remains to consider $a > \pi$.

For $B = b_n/\varepsilon > 0$, the elementary strict ceiling inequality gives

$$M_n < \sqrt{B^2 + \frac{1}{4}} + \frac{1}{2}.$$

Both sides are nonnegative, and $\sqrt{B^2 + 1/4} \leq B + 1/2$. Squaring and restoring the scale yields

$$\varepsilon^2 M_n^2 < b_n^2 + \varepsilon b_n + \varepsilon^2. \quad (3.13)$$

The function $x \mapsto \sqrt{a^2 - \pi^2 x^2}$ is strictly decreasing on its positive interval. Hence

$$\sum_{n=1}^N b_n < \int_0^N \sqrt{a^2 - \pi^2 x^2} dx \leq \int_0^{a/\pi} \sqrt{a^2 - \pi^2 x^2} dx = \frac{a^2}{4}. \quad (3.14)$$

Also $N < a/\pi$, including every radial wall because the wall level is excluded. Summation of (3.13) therefore gives

$$\varepsilon^2 \mathcal{P}_\varepsilon(a) < S_0(a) + \frac{\varepsilon a^2}{4} + \frac{\varepsilon^2 a}{\pi}. \quad (3.15)$$

It is enough to prove

$$D(a) \geq \varepsilon \left(I_0(a) + \frac{a^2}{4} \right) + \frac{\varepsilon^2 a}{\pi}, \quad (3.16)$$

because then (3.15) is at most $I_0(a)(1 - \varepsilon)$, and hence is strictly less than $I_0(a)m_\varepsilon$.

After division by a^2 , the right-hand side of (3.16) is

$$\frac{2\varepsilon a}{3\pi} + \frac{\varepsilon}{4} + \frac{\varepsilon^2}{\pi a}.$$

The low-piece inequalities $a \leq \pi/(4\varepsilon)$ and $a \geq \pi$, together with the elementary strict bound $\pi > 3$, imply

$$\frac{2\varepsilon a}{3\pi} + \frac{\varepsilon}{4} + \frac{\varepsilon^2}{\pi a} \leq \frac{1}{6} + \frac{\varepsilon}{4} + \frac{\varepsilon^2}{9}. \quad (3.17)$$

The last expression increases with $\varepsilon > 0$. On the closed face $\varepsilon = 1/8$, its exact reserve below $2/5$ is

$$\frac{2}{5} - \left(\frac{1}{6} + \frac{1}{32} + \frac{1}{576} \right) = \frac{577}{2880} > 0. \quad (3.18)$$

Thus Lemma 3.2 proves (3.16), and Lemma 3.1 together with (3.15) proves (3.12).

All inequalities allow $a = \pi/(4\varepsilon)$. Since $\varepsilon \leq 1/8$, this face has $a \geq 2\pi$. At its corner $(\varepsilon, a) = (1/8, 2\pi)$, the strict radial convention is $(N, t) = (1, 1)$, and the positive reserve (3.18) remains unchanged. Thus the low proof owns the whole common face. $\square$

3.2 The complementary-action comparison

For $r \in \{\rho, 1\}$ and $0 \leq y \leq ar$, define

$$J_r(y) = \sqrt{a^2 r^2 - y^2} - y \arccos \frac{y}{ar}, \quad (3.19)$$

and set

$$\mathcal{A}(y) = \begin{cases} \frac{J_1(y) - J_\rho(y)}{\pi\varepsilon}, & 0 \leq y \leq \rho a, \\ \frac{J_1(y)}{\pi\varepsilon}, & \rho a \leq y \leq a. \end{cases} \quad (3.20)$$

We extend $\mathcal{A}(y) = 0$ for $y > a$.

Lemma 3.4 (Action geometry and curvature switch). *The function $\mathcal{A}$ is continuous, continuously differentiable at $y = \rho a$, and strictly decreasing from*

$$T := \mathcal{A}(0) = \frac{a}{\pi} \quad \text{to} \quad \mathcal{A}(a) = 0.$$

Let $R : [0, T] \rightarrow [0, a]$ be its decreasing inverse, let

$$\tau = \mathcal{A}(\rho a), \quad F(t) = R(t)^2, \quad g(t) = -F'(t).$$

Then

$$\begin{aligned} g &\text{ decreases on } (0, \tau), & g &\text{ increases on } (\tau, T), \\ F(0) &= a^2, & F(\tau) &= \rho^2 a^2, & F(T) &= 0, & g(T) &= 2\pi\rho a, \end{aligned} \quad (3.21)$$

where $g(T)$ denotes the one-sided limit. Moreover $g(t) = O(t^{-1/3})$ as $t \downarrow 0$.

Proof. Direct differentiation on the appropriate open intervals gives

$$J'_r(y) = -\arccos \frac{y}{ar}, \quad J''_r(y) = \frac{1}{\sqrt{a^2 r^2 - y^2}}. \quad (3.22)$$

At $y = \rho a$, $J_\rho(\rho a) = 0$, and the two one-sided first derivatives of (3.20) both equal $-\arccos \rho/(\pi\varepsilon)$. Thus the two pieces and their first derivatives match. Both derivatives are negative away from $y = a$, so the inverse exists on exactly the stated interval.

On the outer branch $\rho a < y < a$, put $\alpha = \arccos(y/a)$. If $t = \mathcal{A}(y)$, then

$$g(t) = -\frac{2y}{\mathcal{A}'(y)} = \frac{2\pi\varepsilon y}{\alpha}. \quad (3.23)$$

The function y/α increases with y , whereas $y = R(t)$ decreases with t ; hence g decreases on $0 < t < \tau$.

On the inner branch $0 < y < \rho a$, put

$$\delta(y) = \arccos \frac{y}{a} - \arccos \frac{y}{\rho a} > 0.$$

Differentiating $\arccos(y/(ar))$ with respect to r gives the exact identity

$$\frac{\delta(y)}{y} = \int_\rho^1 \frac{dr}{r\sqrt{a^2 r^2 - y^2}}. \quad (3.24)$$

The integrand increases strictly with y , so $\delta(y)/y$ increases, and $y/\delta(y)$ decreases, with y . Since

$$g(t) = \frac{2\pi\varepsilon y}{\delta(y)}$$

and $y = R(t)$ decreases, g increases on $\tau < t < T$. Formula (3.23) and its inner counterpart agree at the interface.

The three values of F follow from $R(0) = a$, $R(\tau) = \rho a$, and $R(T) = 0$. As $y \downarrow 0$, (3.24) gives

$$\delta(y) = \frac{\varepsilon}{\rho a} y + O(y^3),$$

and therefore $g(T) = 2\pi\rho a$.

Finally, on the outer branch write $y = a \cos \theta$. Then

$$t = \frac{a}{\pi\varepsilon}(\sin \theta - \theta \cos \theta) \sim \frac{a\theta^3}{3\pi\varepsilon}, \quad g(t) = \frac{2\pi\varepsilon a \cos \theta}{\theta} = O(t^{-1/3}). \quad (3.25)$$

This proves the asserted improper endpoint behavior. All square roots and arccosines in the proof are taken on their stated closed domains; every division occurs in an open interval where its denominator is positive, and the endpoint zeros are handled by the limits just computed. $\square$

Lemma 3.5 (Strict phase-to-action layer cake). *For $\ell \in \mathbb{N}_0$, set*

$$y_\ell = \varepsilon \left(\ell + \frac{1}{2} \right), \quad q_\ell = \left[\mathcal{A}(y_\ell) + \frac{1}{4} \right]_<, \quad P_{\mathcal{A}} = \varepsilon \sum_{\ell \geq 0} y_\ell q_\ell. \quad (3.26)$$

Then

$$N_D(A_{\rho,1}, K^2) \leq \frac{2P_{\mathcal{A}}}{\varepsilon^2}. \quad (3.27)$$

For $n \geq 1$, put $x_n = n - \frac{1}{4}$, and define

$$M_\varepsilon(x) = \# \left\{ \ell \in \mathbb{N}_0 : \varepsilon \left(\ell + \frac{1}{2} \right) < x \right\} = \max \left\{ 0, \left\lceil \frac{x}{\varepsilon} - \frac{1}{2} \right\rceil \right\}. \quad (3.28)$$

The proxy has the exact finite representation

$$P_{\mathcal{A}} = \frac{\varepsilon^2}{2} \sum_{\substack{n \geq 1 \\ x_n < T}} M_\varepsilon(R(x_n))^2. \quad (3.29)$$

Moreover,

$$I := \frac{1}{2} \int_0^T F(t) dt = \frac{a^3}{3\pi} \left(1 - \varepsilon + \frac{\varepsilon^2}{3} \right), \quad \frac{2I}{\varepsilon^2} = \frac{2}{9\pi} (1 - \rho^3) K^3. \quad (3.30)$$

Proof. For $y_\ell < a$, equations (3.3) and (3.4) show that the number of radial levels in channel ℓ is at most q_ℓ . If $y_\ell \geq a$, equivalently $\nu_\ell \geq K$, the product comparison gives

$$\lambda_{\ell,1} \geq \frac{\pi^2}{\varepsilon^2} + \ell(\ell+1) = \frac{\pi^2}{\varepsilon^2} + \nu_\ell^2 - \frac{1}{4} > K^2,$$

so there is no radial level; at $y_\ell = a$, the definition also gives $q_\ell = [1/4]_< = 0$. Multiplicity $2\ell + 1 = 2y_\ell/\varepsilon$ now yields

$$N_D(A_{\rho,1}, K^2) \leq \sum_{\ell \geq 0} \frac{2y_\ell}{\varepsilon} q_\ell = \frac{2P_{\mathcal{A}}}{\varepsilon^2},$$

which proves (3.27). If $\mathcal{A}(y_\ell) + 1/4 = m \in \mathbb{Z}$, then $q_\ell = m - 1$, so the phase-proxy wall is excluded exactly as required.

For every $y \geq 0$, the strict bracket is

$$\left[\mathcal{A}(y) + \frac{1}{4} \right]_< = \# \{ n \geq 1 : x_n < \mathcal{A}(y) \}.$$

The sums are finite. Since $\mathcal{A}$ and R are decreasing, $x_n < \mathcal{A}(y_\ell)$ is equivalent to $y_\ell < R(x_n)$. Therefore

$$\begin{aligned} P_{\mathcal{A}} &= \varepsilon \sum_{\substack{n \geq 1 \\ x_n < T}} \sum_{y_\ell < R(x_n)} y_\ell \\ &= \varepsilon \sum_{\substack{n \geq 1 \\ x_n < T}} \varepsilon \sum_{\ell=0}^{M_\varepsilon(R(x_n))-1} \left(\ell + \frac{1}{2} \right) \\ &= \frac{\varepsilon^2}{2} \sum_{\substack{n \geq 1 \\ x_n < T}} M_\varepsilon(R(x_n))^2, \end{aligned}$$

because $\sum_{\ell=0}^{M-1}(\ell + 1/2) = M^2/2$. This proves (3.29). Formula (3.28) follows from the same strict counting; at $x/\varepsilon = m + 1/2$, it equals m , excluding $\ell = m$.

Differentiation of (3.19) with respect to r gives

$$\frac{\partial J_r(y)}{\partial r} = \sqrt{a^2 - \frac{y^2}{r^2}} \quad (0 \leq y \leq ar).$$

Consequently the two cases of (3.20) combine as

$$\mathcal{A}(y) = \frac{1}{\pi\varepsilon} \int_{\rho}^1 \left(a^2 - \frac{y^2}{r^2} \right)_+^{1/2} dr, \quad (3.31)$$

where $s_+ = \max\{s, 0\}$. The area identity for a decreasing inverse and Tonelli's theorem give

$$\frac{1}{2} \int_0^T R(t)^2 dt = \int_0^a y \mathcal{A}(y) dy.$$

Using (3.31) and then $y = rz$,

$$\begin{aligned} \int_0^a y \mathcal{A}(y) dy &= \frac{1}{\pi\varepsilon} \int_{\rho}^1 r^2 dr \int_0^a z \sqrt{a^2 - z^2} dz \\ &= \frac{a^3}{3\pi\varepsilon} \int_{\rho}^1 r^2 dr = \frac{a^3}{3\pi} \left(1 - \varepsilon + \frac{\varepsilon^2}{3} \right). \end{aligned}$$

This is (3.30); its final equality follows from $1 - \rho^3 = 3\varepsilon m_\varepsilon$ and $K = a/\varepsilon$. $\square$

Lemma 3.6 (Shifted sawtooth and radial quadrature deficit). *Let*

$$C(t) = \#\{n \geq 1 : n - \frac{1}{4} < t\}, \quad w(t) = t - C(t), \quad \mathfrak{W}(t) = \int_0^t w(s) ds,$$

and put $\Psi(t) = \mathfrak{W}(t) - t/4$. Then, for all $t \geq 0$,

$$\mathfrak{W}(t) \geq 0, \quad -\frac{1}{32} \leq \Psi(t) \leq \frac{3}{32}. \quad (3.32)$$

Furthermore, with

$$D_{\text{rad}} = \int_0^T F(t) dt - \sum_{\substack{n \geq 1 \\ x_n < T}} F(x_n), \quad (3.33)$$

one has

$$D_{\text{rad}} \geq \frac{\rho^2 a^2}{4} - \frac{\pi \rho a}{4}. \quad (3.34)$$

Proof. On $0 \leq t \leq 3/4$, $C(t) = 0$, and hence

$$\Psi(t) = \frac{1}{2} \left(t - \frac{1}{4} \right)^2 - \frac{1}{32}. \quad (3.35)$$

For $x_n = n - 1/4$, $n \geq 1$, and $t = x_n + u$ with $0 \leq u \leq 1$, direct integration gives

$$\Psi(t) = \frac{1}{2} \left(u - \frac{1}{2} \right)^2 - \frac{1}{32}. \quad (3.36)$$

These formulas agree at cell endpoints and prove the two-sided bound for Ψ . On the first cell $\mathfrak{W} = t^2/2 \geq 0$. Thereafter $\mathfrak{W} = t/4 + \Psi \geq t/4 - 1/32 > 0$, proving (3.32). At a jump

x_n , strict counting gives the one-sided values $w(x_n^-) = 3/4$ and $w(x_n^+) = -1/4$, while $\mathfrak{W}$ and Ψ are continuous.

Distributionally, $dw = dt - dC$. Since $F(T) = 0$, an atom at $x_n = T$ has zero weight and is consistent with its exclusion from (3.33). Stieltjes integration by parts, first on $[\eta, T]$ and then with $\eta \downarrow 0$, gives

$$D_{\text{rad}} = \int_0^T w(t)g(t) dt. \quad (3.37)$$

Indeed, (3.25), together with $w(t) = t$ and $\mathfrak{W}(t) = t^2/2$ near zero, gives $w(t)g(t) = O(t^{2/3})$ and $\mathfrak{W}(t)g(t) = O(t^{5/3}) \rightarrow 0$, so the improper boundary term vanishes.

Split (3.37) at the actual value τ , with no assumption that τ belongs to any particular sawtooth cell. On $(0, \tau)$, g decreases and $\mathfrak{W} \geq 0$; hence

$$\int_0^\tau wg = \mathfrak{W}(\tau)g(\tau) - \int_0^\tau \mathfrak{W} dg \geq \mathfrak{W}(\tau)g(\tau). \quad (3.38)$$

On (τ, T) , g increases and $w = 1/4 + \Psi'$ almost everywhere. Adding its integration-by-parts identity to (3.38), and using $\mathfrak{W}(\tau) = \tau/4 + \Psi(\tau)$, cancels the interface value and yields

$$D_{\text{rad}} \geq \frac{F(\tau)}{4} + \frac{\tau g(\tau)}{4} + \Psi(T)g(T) - \int_\tau^T \Psi dg.$$

Because dg is a positive measure on (τ, T) , the bounds (3.32) imply

$$\begin{aligned} D_{\text{rad}} &\geq \frac{F(\tau)}{4} - \frac{g(T)}{8} + g(\tau) \left(\frac{\tau}{4} + \frac{3}{32} \right) \\ &\geq \frac{F(\tau)}{4} - \frac{g(T)}{8}. \end{aligned}$$

The two discarded quantities, $-\int_0^\tau \mathfrak{W} dg$ and $g(\tau)(\tau/4 + 3/32)$, are explicitly nonnegative. Substitution of $F(\tau) = \rho^2 a^2$ and $g(T) = 2\pi\rho a$ from Lemma 3.4 proves (3.34). Continuity of $\mathfrak{W}, \Psi, g$ at the split shows that the same proof covers τ below, on, or above every sawtooth grid point. $\square$

Proposition 3.7 (The inclusive high-action piece). *If*

$$0 < \varepsilon \leq \frac{1}{8}, \quad a \geq \frac{\pi}{4\varepsilon},$$

then

$$N_D(A_{\rho,1}, K^2) < \frac{2}{9\pi}(1 - \rho^3)K^3. \quad (3.39)$$

Proof. The number of active radial action layers is

$$\begin{aligned} J_{\text{rad}} &:= \#\{n \geq 1 : x_n < T\} = \left\lceil T + \frac{1}{4} \right\rceil - 1 \\ &< T + \frac{1}{4} = \frac{a}{\pi} + \frac{1}{4}. \end{aligned} \quad (3.40)$$

At a wall $x_n = T$, this gives $J_{\text{rad}} = n - 1$, so the equality layer is excluded.

For every included $x_n < T$, let $x = R(x_n) > 0$. The exact count (3.28) satisfies

$$M_\varepsilon(x) < \frac{x}{\varepsilon} + \frac{1}{2}.$$

The two sides are nonnegative, so after squaring and using $x \leq a$,

$$\varepsilon^2 M_\varepsilon(x)^2 < x^2 + \varepsilon x + \frac{\varepsilon^2}{4} \leq x^2 + \varepsilon a + \frac{\varepsilon^2}{4}. \quad (3.41)$$

At a half-integer wall $x/\varepsilon = m + 1/2$, the strict count is m , so (3.41) remains strict. Summing and using (3.40) gives

$$\varepsilon^2 \sum_{\substack{n \geq 1 \\ x_n < T}} M_\varepsilon(R(x_n))^2 < \sum_{\substack{n \geq 1 \\ x_n < T}} F(x_n) + E, \quad (3.42)$$

where

$$E = \left(\frac{a}{\pi} + \frac{1}{4} \right) \left(\varepsilon a + \frac{\varepsilon^2}{4} \right). \quad (3.43)$$

By Lemma 3.6 and $a \geq \pi/(4\varepsilon)$,

$$\frac{D_{\text{rad}}}{a^2} \geq \frac{\rho^2}{4} - \frac{\pi\rho}{4a} \geq \frac{(1-\varepsilon)(1-5\varepsilon)}{4}. \quad (3.44)$$

The last expression has derivative $(-6 + 10\varepsilon)/4 < 0$ on $0 < \varepsilon \leq 1/8$, and therefore

$$\frac{D_{\text{rad}}}{a^2} \geq \frac{21}{256}. \quad (3.45)$$

On the other hand, expansion of (3.43) and $1/a \leq 4\varepsilon/\pi$ give

$$\frac{E}{a^2} \leq \frac{\varepsilon}{\pi} + \frac{\varepsilon^2}{\pi} + \frac{\varepsilon^3}{\pi^2} + \frac{\varepsilon^4}{\pi^2}. \quad (3.46)$$

This scaling is deliberately non-strict: equality occurs at the common optical face $a = \pi/(4\varepsilon)$. The right-hand side is strictly increasing in $\varepsilon > 0$. At $\varepsilon = 1/8$, the elementary strict inequality $\pi > 3$ gives

$$\frac{E}{a^2} \leq \frac{1}{8\pi} + \frac{1}{64\pi} + \frac{1}{512\pi^2} + \frac{1}{4096\pi^2} < \frac{193}{4096}. \quad (3.47)$$

Combining (3.45) and (3.47) gives the exact positive reserve

$$D_{\text{rad}} - E > a^2 \left(\frac{21}{256} - \frac{193}{4096} \right) = \frac{143}{4096} a^2 > 0. \quad (3.48)$$

Equations (3.33), (3.42), and (3.48) now give the complete strict chain

$$\varepsilon^2 \sum_{\substack{n \geq 1 \\ x_n < T}} M_\varepsilon(R(x_n))^2 < \sum_{\substack{n \geq 1 \\ x_n < T}} F(x_n) + E < \int_0^T F(t) dt. \quad (3.49)$$

By Lemma 3.5, this is $P_{\mathcal{A}} < I$, and hence

$$N_D(A_{\rho,1}, K^2) \leq \frac{2P_{\mathcal{A}}}{\varepsilon^2} < \frac{2I}{\varepsilon^2} = \frac{2}{9\pi} (1 - \rho^3) K^3.$$

Every inequality used above includes $a = \pi/(4\varepsilon)$. At that face the non-strict scalings in (3.44)–(3.46) are equalities, but the strict estimate $\pi > 3$ retains the reserve (3.48). Thus the high proof independently owns the whole common face, including $(\varepsilon, a) = (1/8, 2\pi)$. $\square$

Lemma 3.8 (Thin-shell endpoint). *For $7/8 \leq \rho < 1$ and $K \geq 0$,*

$$N_D(A_{\rho,1}, K^2) \leq \frac{2}{9\pi} (1 - \rho^3) K^3. \quad (3.50)$$

Equality holds at $K = 0$, and the displayed inequality is strict when $K > 0$.

Proof. At $K = 0$, positivity of the Dirichlet operator gives a zero strict count, and the right-hand side is also zero. Suppose $K > 0$. Then $0 < \varepsilon \leq 1/8$ and $a > 0$. The two inclusive intervals

$$0 \leq a \leq \frac{\pi}{4\varepsilon}, \quad a \geq \frac{\pi}{4\varepsilon}$$

cover every $a \geq 0$, with no gap and no third range. The first is proved by Proposition 3.3; the second is proved by Proposition 3.7. Both propositions include their common face. Formula (3.2) identifies the low-piece target with the right-hand side of (3.50). This completes the proof on the closed face $\rho = 7/8$ and for every $7/8 < \rho < 1$. The limit $\rho \uparrow 1$ means $\varepsilon \downarrow 0$ through positive values; the nonexistent shell $\varepsilon = 0$ is never inserted into a denominator or treated as part of the domain. $\square$

Remark 3.9 (Dependency boundary). No finite list of Bessel zeros, no numerical root certificate, and no lower bound for a first Bessel zero is used in the proof of Lemma 3.8. In particular, the inequalities of Lorch for $j_{\nu,1}$ are dependencies of the compact-middle finite-channel staircase, not of the endpoint $7/8 \leq \rho < 1$. The only non-elementary input specific to the high piece is the all-domain optimized phase estimate (3.3), whose proof appears earlier in this manuscript; all product, action, inverse, curvature, Stieltjes, rounding, seam, and constant estimates used after that input have been proved above.

4 The compact-middle argument in full detail

We retain the strict counting convention $[x]_{<} := \#\{n \in \mathbb{N} : n < x\}$, put

$$W(\rho, K) := \frac{2}{9\pi}(1 - \rho^3)K^3, \quad z_\rho := \frac{\pi}{1 - \rho}, \quad k_m(\rho) := \sqrt{z_\rho^2 + m(m+1)},$$

and denote by $q_{\ell,n}(\rho)$ the n -th positive frequency in angular channel ℓ . Its multiplicity is $2\ell + 1$. The phase reduction proved earlier gives

$$N_D(A_{\rho,1}, K^2) \leq P_{\text{coarse}}(\rho, K) := \sum_{\ell+1/2 < K} (2\ell + 1) \left[A_{\rho,K}(\ell + 1/2) + \frac{1}{4} \right]_{<}, \quad (4.1)$$

where $A_{\rho,K} = G_K - G_{\rho K}$ and

$$G_a(\nu) = \begin{cases} \frac{1}{\pi} \left(\sqrt{a^2 - \nu^2} - \nu \arccos \frac{\nu}{a} \right), & 0 \leq \nu < a, \\ 0, & \nu \geq a. \end{cases}$$

The value at both interfaces is therefore fixed by zero extension. This will matter at $\nu = \rho K$, at $\nu = K$, and at every integer proxy wall.

4.1 Constants and the primitive analytic envelope

For $0 \leq s \leq 1$, set

$$G_1(s) = \frac{1}{\pi} \left(\sqrt{1 - s^2} - s \arccos s \right), \quad \omega_0 = G_1(1/2) = \frac{\sqrt{3}}{2\pi} - \frac{1}{6},$$

$$C_* = \frac{1}{2} + \frac{8\sqrt{2}}{15}, \quad C_0 = 1 + \frac{8\sqrt{2}}{15}, \quad \rho_* := \frac{\omega_0}{1 + 2C_*} = \frac{\omega_0}{2C_0}.$$

For $0 < \rho < 1$, define

$$a_\rho = \frac{2\pi\rho}{1 - \rho}, \quad \eta_\rho = G_1(\max\{\rho, 1/2\}),$$

$$K_0(\rho) = \left(\frac{\sqrt{a_\rho} + \sqrt{a_\rho + 4\eta_\rho C_0}}{2\eta_\rho} \right)^2, \quad H_0(\rho) = \frac{C_*}{\omega_0 - \rho} \quad (\rho < \omega_0). \quad (4.2)$$

We first re-establish every primitive owner that is needed to form the compact residual. This prevents the residual description from hiding an unproved coverage assumption.

Proposition 4.1 (Primitive analytic owners). *The Pólya bound holds on each of the following inclusive domains:*

1. $2\rho K \leq 1$;
2. $(1 - \rho)K \leq \pi$;
3. $0 < \rho < \omega_0$ and $K(\omega_0 - \rho) \geq C_*$, equivalently $K \geq H_0(\rho)$;
4. $0 < \rho \leq \rho_*$, for every $K \geq 0$;
5. $K \geq K_0(\rho)$;
6. $0 < 1 - \rho \leq 1/6$ and $K(1 - \rho)^2 \geq 54$;
7. $7/8 \leq \rho < 1$, for every $K \geq 0$.

In particular, on the full ratio interval $0 < \rho < 1$, the bound holds whenever $K \geq 295^2$.

Proof. For item 1, $\mu = \rho K \leq 1/2$; every shifted half-integer tail begins at or above the inner interface and is controlled by the convex high-angular tail inequality. Summing the tail integrals with the weighted identity gives $N_D \leq W$. For item 2, $K \leq z_\rho = q_{0,1}$, and (4.14) (whose elementary proof is repeated below) shows that no radial mode is strictly below K .

We record the tail bookkeeping used in items 3–5. Put

$$x_r = r + \frac{1}{2}, \quad q_\ell = \left\lfloor A_{\rho,K} \left(\ell + \frac{1}{2} \right) + \frac{1}{4} \right\rfloor$$

and

$$\mathcal{T}_r = \left\lfloor A_{\rho,K}(x_r) + \frac{1}{4} \right\rfloor + 2 \sum_{j \geq 1} \left\lfloor A_{\rho,K}(x_r + j) + \frac{1}{4} \right\rfloor.$$

All sums are finite because $A_{\rho,K} = 0$ at and beyond K . For integers $a < b$, write

$$\mathbf{T}(h, a, b) = \frac{1}{2} \lfloor h(a) \rfloor + \sum_{j=a+1}^{b-1} \lfloor h(j) \rfloor + \frac{1}{2} \lfloor h(b) \rfloor.$$

The elementary multiplicity identity

$$\sum_{\ell \geq 0} (2\ell + 1) q_\ell = \sum_{r \geq 0} \left(q_r + 2 \sum_{\ell > r} q_\ell \right) = \sum_{r \geq 0} \mathcal{T}_r$$

shows that $P_{\text{coarse}} \leq \sum_r \mathcal{T}_r$. If every active tail satisfies

$$\mathcal{T}_r \leq 2 \int_{x_r}^K A_{\rho,K}(z) dz,$$

then, with $f_3(z) = \#\{r \geq 0 : x_r \leq z\}$ and $F_3(z) = \int_0^z f_3(t) dt$,

$$P_{\text{coarse}} \leq 2 \int_0^K f_3(z) A_{\rho,K}(z) dz.$$

Writing $z = m + 1/2 + s$, $m \in \mathbb{N}_0$, $0 \leq s < 1$, gives

$$\frac{z^2}{2} - F_3(z) = \frac{1}{2} \left(s - \frac{1}{2} \right)^2 \geq 0;$$

for $0 \leq z < 1/2$, the same inequality is immediate. Since $A_{\rho,K}$ is decreasing, absolutely continuous, and vanishes at K , integration by parts yields

$$P_{\text{coarse}} \leq - \int_0^K z^2 A'_{\rho,K}(z) dz = \int_0^K 2z A_{\rho,K}(z) dz = \frac{2}{9\pi} (1 - \rho^3) K^3 = W(\rho, K).$$

For the final equality, scaling reduces each G_a -integral to

$$\int_0^1 2x G_1(x) dx = \frac{2}{\pi} \left(\frac{1}{3} - \frac{2}{9} \right) = \frac{2}{9\pi}.$$

We shall use the exact trapezoidal-floor estimates proved in the one-dimensional reduction. For a nonnegative concave h , $\mathbf{T}(h, a, b) \leq \int_a^b h(t) dt$. If g is nonnegative, decreasing, convex, and $1/2$ -Lipschitz, vanishes at b , and is $1/3$ -Lipschitz after t_* , then

$$\mathbf{T}(g + 1/4, a, b) \leq \int_a^b g(t) dt - \frac{1}{4} \lfloor g(t_*) \rfloor.$$

The corresponding shifted convex estimate gives $\mathcal{T}_r \leq 2 \int_{x_r}^K G_K(z) dz$ whenever $x_r \geq \mu$. Indeed, on that interval $A = G_K$, and

$$G'_K(z) = -\frac{1}{\pi} \arccos \frac{z}{K} \in [-1/2, 0], \quad G''_K(z) = \frac{1}{\pi \sqrt{K^2 - z^2}} > 0.$$

We now prove item 3. Write $A = A_{\rho,K}$. Only a start $x_0 < \mu = \rho K$ remains. Put

$$n = \lfloor \mu - x_0 \rfloor, \quad q = x_0 + n, \quad B = \lceil K - x_0 \rceil.$$

Thus $q \leq \mu < q + 1$, $n < B$, and zero extension gives $G_K(x_0 + B) = 0$. The action is concave on $[x_0, q]$, because

$$A''(z) = \frac{1}{\pi \sqrt{K^2 - z^2}} - \frac{1}{\pi \sqrt{\mu^2 - z^2}} < 0 \quad (0 < z < \mu).$$

Moreover $q \leq \mu = \rho K < \omega_0 K < K/2$. Splitting at n , using the concave estimate on the head and the improved convex estimate on the G_K -suffix with $t_* = K/2 - x_0$, gives, also when $n = 0$,

$$\frac{\mathcal{T}_r}{2} \leq \int_{x_0}^q A(z) dz + \frac{n}{4} + \int_q^K G_K(z) dz - \frac{1}{4} \lfloor \omega_0 K \rfloor.$$

The only integral mismatch is

$$\delta := \int_q^\mu G_\mu(z) dz.$$

To bound it, write $t = \cos \theta$. Concavity of $\sin$ on $[0, \pi/4]$ gives $\sin(\theta/2) \geq \sqrt{2}\theta/\pi$, hence $\arccos t \leq (\pi/2)\sqrt{1-t}$ for $0 \leq t \leq 1$. Since

$$G_\mu(z) = \frac{1}{\pi} \int_z^\mu \arccos \frac{t}{\mu} dt,$$

we obtain

$$0 \leq \delta \leq \frac{2(\mu - q)^{5/2}}{15\sqrt{\mu}} < \frac{2\sqrt{2}}{15},$$

where a low tail has $\mu > x_0 \geq 1/2$ and $0 \leq \mu - q < 1$. Consequently

$$\frac{\mathcal{T}_r}{2} \leq \int_{x_0}^K A(z) dz + \delta - \frac{[\omega_0 K] - n}{4}.$$

Finally $n \leq \rho K - 1/2$ and $[y] > y - 1$, so

$$[\omega_0 K] - n > K(\omega_0 - \rho) - \frac{1}{2} \geq C_* - \frac{1}{2} = \frac{8\sqrt{2}}{15} > 4\delta.$$

Thus every low-interface tail is strict, every high-interface tail is non-strict, and the weighted scaffold proves item 3, including the face $K(\omega_0 - \rho) = C_*$.

For item 4, if $\mu \leq 1/2$ use item 1. Otherwise $K > 1/(2\rho)$. Because $\rho_* = \omega_0/(1 + 2C_*)$, for $0 < \rho \leq \rho_*$,

$$K(\omega_0 - \rho) > \frac{\omega_0 - \rho}{2\rho} \geq \frac{\omega_0 - \rho_*}{2\rho_*} = C_*,$$

and item 3 applies. All faces are included; at $K = 0$ both sides vanish.

For item 5, fix a low-interface tail $x_0 = r + 1/2 < \mu$, retain the preceding definition of $\mathcal{T}_r$, and write $A = A_{\rho, K}$. Put $n = \lfloor \mu - x_0 \rfloor$, $q = x_0 + n$, and $B = \lceil K - x_0 \rceil$. On $[0, n]$, $h(t) = A(x_0 + t) + 1/4$ is decreasing, concave, and Lipschitz with constant $c_\rho = \arccos \rho/\pi$. If p is the last index of its first constant ordinary-floor shelf (put $p = n$ when the shelf never drops), the concave trapezoidal-floor inequality gives

$$\mathbf{T}(h, 0, n) \leq \int_0^n A(x_0 + t) dt + \frac{n}{4} - \frac{1 - c_\rho}{2}(n - p). \quad (4.3)$$

For $n = 0$, the head is absent and $p = 0$.

Let $\vartheta = -A'$. Direct differentiation gives

$$\vartheta'(z) = \frac{1}{\pi\sqrt{\mu^2 - z^2}} - \frac{1}{\pi\sqrt{K^2 - z^2}} \geq \frac{1 - \rho}{\pi\rho K} =: \kappa_\rho.$$

Equal floors on the first shelf imply $0 \leq A(x_0) - A(x_0 + p) < 1$, including at an integer endpoint. Hence

$$1 > \int_{x_0}^{x_0+p} \vartheta(z) dz \geq \frac{\kappa_\rho p^2}{2}, \quad p < \sqrt{a_\rho K}.$$

For the convex tail set $s = \max\{q, K/2\}$. The function $G_K(x_0 + t)$ is decreasing and convex, is Lipschitz $1/2$ on the whole tail and $1/3$ after s , and $G_K(s) \geq K\eta_\rho$. The convex trapezoidal-floor inequality therefore gives

$$\mathbf{T}(G_K(x_0 + \cdot) + 1/4, n, B) \leq \int_n^B G_K(x_0 + t) dt - \frac{1}{4}[K\eta_\rho]. \quad (4.4)$$

At $j = n$, the two half endpoint weights in (4.3)–(4.4) majorize the original full A -sample. Their only integral mismatch is

$$\delta = \int_q^\mu G_\mu(z) dz, \quad 0 \leq \delta \leq \frac{2(\mu - q)^{5/2}}{15\sqrt{\mu}} < \frac{2\sqrt{2}}{15}.$$

Writing $d_\rho = 1 - 2c_\rho = 2 \arcsin \rho/\pi$, combination of the head and tail yields

$$\frac{\mathcal{T}_r}{2} \leq \int_{x_0}^K A(z) dz + \delta - \frac{M}{4}, \quad M = [K\eta_\rho] + d_\rho(n - p) - p.$$

Since $d_\rho > 0$ and $p < \sqrt{a_\rho K}$, the strict floor inequality, valid even when $K\eta_\rho$ is an integer, gives

$$M > K\eta_\rho - 1 - \sqrt{a_\rho K}.$$

Thus the exact sufficient condition is

$$K\eta_\rho - \sqrt{a_\rho K} \geq C_0. \quad (4.5)$$

Indeed it implies $M > 8\sqrt{2}/15 > 4\delta$, so every low-interface tail is strictly below twice its action integral. Tails beginning at or above μ satisfy the non-strict convex high-interface bound; the weighted identity completes the proof. The cases $n = 0$, $p = 0$, $p = n$, $q = \mu$, q on either side of $K/2$, and all strict-floor walls are included in the displayed inequalities.

Writing $Y = \sqrt{K}$, the left side minus C_0 is $\eta_\rho Y^2 - \sqrt{a_\rho} Y - C_0$. It has one positive root, and the square of that root is precisely $K_0(\rho)$ in (4.2). Hence (4.5) holds, including equality, exactly for $K \geq K_0(\rho)$.

We give the seam proof for item 6 because it is the nontrivial owner in the primitive mask. Put

$$\varepsilon = 1 - \rho, \quad y = \sqrt{\varepsilon}, \quad \kappa = K\varepsilon^2, \quad \mu = \rho K, \quad c = \frac{\arccos \rho}{\pi}, \quad d = 1 - 2c, \quad \eta = G_1(\rho).$$

For a shifted low-interface tail beginning at $x_0 = r_0 + 1/2 < \mu$, write

$$n = \lfloor \mu - x_0 \rfloor, \quad q = x_0 + n, \quad \beta = \mu - q, \quad h_j = \left\lfloor A(x_0 + j) + \frac{1}{4} \right\rfloor.$$

Let p be the last index of the first constant plateau, set $m = n - p$, $R = p - dm$, and use $p = m = R = 0$ if $n = 0$. The exact one-tail decomposition is

$$\frac{\mathcal{T}_{r_0}}{2} \leq \int_{x_0}^K A(z) dz + \delta - \frac{M}{4}, \quad M = \lfloor K\eta \rfloor - R, \quad 0 \leq \delta < \frac{2\sqrt{2}}{15}. \quad (4.6)$$

The curvature shelf estimate gives

$$p < \sqrt{\frac{2\pi\rho K}{\varepsilon}}. \quad (4.7)$$

It remains to prove $M > 4\delta$ on $0 < y \leq 1/\sqrt{6}$, $\kappa \geq 54$.

Set $P = py$, $r = Ry$, and $S = (p+m)y$. The elementary half-angle estimate $\cos(3\pi/16) < 5/6 \leq \rho$ gives

$$d > 5/8, \quad R > 0 \implies m < \frac{8}{5}p, \quad S < \frac{13}{5}P. \quad (4.8)$$

With $t = x_0/(\rho K)$, $w = 1 - t$, equations (4.7)–(4.8) imply

$$w < \hat{q} := \frac{y^4 + y^3 S}{\kappa \rho} < \frac{1}{1620} + \frac{13}{5} \frac{763}{5000} = \frac{804689}{2025000} < \frac{159}{400}.$$

The estimates use only $\pi < 1571/500$, $140/99 < \sqrt{2} < 99/70$, and $y < 49/120$, each proved by positive rational squaring. In particular, $x_0/K > 241/480 > 1/2$.

For $0 < z < \mu$, let $\sigma(z) = -A'(z)$. Direct differentiation gives

$$\sigma'(z) > 0, \quad \sigma(x_0) \geq \frac{\varepsilon t}{\pi \sqrt{1 - \rho^2 t^2}}.$$

Equal ordinary floors along the plateau imply the strict inequality $A(x_0) - A(x_0 + p) < 1$, even at an integer wall. Put

$$Q = 1 + \frac{y^2}{\kappa} + \frac{yS}{\kappa}.$$

Since $1 - \rho^2 t^2 < 2y^2 Q$, integration of σ across the plateau yields

$$P^2 < \frac{2\pi^2 Q}{(1 - \hat{q})^2}. \quad (4.9)$$

Using $d > 5/8$, replace S by $S_* = (13P - 8r)/5$, and define

$$Q_* = 1 + \frac{y^2}{\kappa} + \frac{yS_*}{\kappa}, \quad q_* = \frac{y^2}{\rho}(Q_* - 1), \quad H(P, r) = \frac{2\pi^2 Q_*}{(1 - q_*)^2}.$$

The right side increases with Q_* , so $P^2 < H(P, r)$. Suppose $r \geq B := 14/3$. Since $R = p - dm \leq p$, one has $0 < r \leq P$. At fixed P , the quantity $S_*(P, r) = (13P - 8r)/5$ decreases with r ; hence Q_* and q_* also decrease. Because $q_* < 159/400 < 1$, the expression $2\pi^2 Q_*/(1 - q_*)^2$ increases with Q_* . Therefore

$$H(P, r) \leq H(P, B).$$

Now freeze $r = B$ and follow the entire segment $B \leq P' \leq P$. The original shelf ceiling remains valid because $P' \leq P$, while

$$0 < S_*(P', B) \leq S_*(P', 0) = \frac{13}{5}P'.$$

Thus the bounds $q_* < 159/400$ and all positive denominators used above remain valid along the path. On it,

$$\partial_{P'} H(P', B) < \frac{16}{5}, \quad \frac{d}{dP'}(P'^2 - H(P', B)) > \frac{92}{15},$$

whereas the initial exact reserve is

$$B^2 - H(B, B) > \frac{2505132463469}{2616573970125} > 0.$$

Integration of the positive derivative yields

$$P^2 - H(P, B) > B^2 - H(B, B) > 0.$$

Consequently $P^2 > H(P, B) \geq H(P, r)$, contradicting $P^2 < H(P, r)$ from (4.9). Hence

$$R < \frac{14}{3\sqrt{\varepsilon}}. \quad (4.10)$$

Finally

$$\eta = \frac{1}{\pi} \int_0^\varepsilon \arccos(1 - v) dv \geq \frac{2\sqrt{2}}{3\pi} \varepsilon^{3/2},$$

so on $\kappa \geq 54$,

$$K\eta > \frac{280000}{17281} \frac{1}{y}.$$

If $R > 0$, the strict inequality $\lfloor x \rfloor > x - 1$, (4.10), and $1/y > 120/49$ give

$$M - 4\delta > \left(\frac{598066}{51843} \right) \frac{120}{49} - 1 - \frac{132}{175} = \frac{80132733}{3024175} > 0.$$

If $R \leq 0$, the stronger reserve is $114694733/3024175 > 0$. Substitution in (4.6) makes every low-interface tail strictly smaller than its action integral; high-interface tails are controlled by convexity. The weighted identity then proves item 6. This argument includes $\kappa =$

54, every plateau wall, $R = 0$, every integer $K\eta$ wall, and the interface $x_0 = \mu$. The derivative cap, initial reserve, and terminal rational payments in this synthetic comparison are regenerated line by line by the ledger printed in Supplement S1, *Standalone exact-rational analytic ledgers*.

Item 7 is the two-piece thin-shell theorem proved by the exact product deficit for low optical frequency and the complementary-action Stieltjes deficit for high optical frequency. For completeness, its two closing reserves are

$$\frac{2}{5} - \left(\frac{1}{6} + \frac{1}{32} + \frac{1}{576} \right) = \frac{577}{2880} > 0, \quad D_{\text{rad}} - E > \frac{143}{4096} a^2 > 0,$$

with $a = (1 - \rho)K$; the common face is included. Thus it supplies the entire closed ratio face $\rho = 7/8$ needed here.

It remains to prove the global frequency stated in the proposition. On $[\rho_*, 1/16]$, item 3 and the increase of H_0 leave only $K < 64$. On $[1/16, 5/6]$, monotonicity of K_0 , proved below, and the exact estimate $K_0(5/6) < 295^2$ leave only $K < 295^2$. On $[5/6, 7/8]$, item 6 leaves only $K < 54/(1 - \rho)^2 \leq 3456$. Item 7 owns every frequency on $[7/8, 1)$. All threshold faces are included, so $K = 295^2$ is covered at every ratio. $\square$

Proposition 4.1 covers the following inclusive sets in $[\rho_*, 7/8] \times [0, \infty)$:

$$\begin{aligned} \rho = \rho_*, \quad \rho = 7/8, \quad 2\rho K \leq 1, \quad (1 - \rho)K \leq \pi, \\ \rho < \omega_0, \quad K(\omega_0 - \rho) \geq C_*, \quad K \geq K_0(\rho), \\ 0 < 1 - \rho \leq 1/6, \quad K(1 - \rho)^2 \geq 54, \quad K \geq 295^2. \end{aligned}$$

Put

$$L(\rho) = \max \left\{ \frac{1}{2\rho}, \frac{\pi}{1 - \rho} \right\},$$

and let $U(\rho)$ be the minimum of all eligible upper owners in (4.2) and the seam owner $54/(1 - \rho)^2$. Since every primitive frequency wall is included, exact complementation gives

$$\mathcal{D}_{16} = \left\{ (\rho, K) : \rho_* < \rho < \frac{7}{8}, \quad L(\rho) < K < U(\rho) \right\}. \quad (4.11)$$

Both frequency inequalities are strict.

We record the switches, since they determine every later face owner. The lower walls meet at

$$\rho_c := \frac{1}{1 + 2\pi}, \quad L(\rho) = \begin{cases} (2\rho)^{-1}, & \rho \leq \rho_c, \\ z_\rho, & \rho \geq \rho_c. \end{cases}$$

On (ρ_*, ω_0) , substituting $K = H_0(\rho)$ into the positive-root equation $\eta_\rho K - \sqrt{a_\rho K} - C_0 = 0$ gives, after a reversible squaring,

$$P(\rho) := (1 - \rho) \left(C_0 \rho - \frac{\omega_0}{2} \right)^2 - 2\pi C_* \rho (\omega_0 - \rho) = 0. \quad (4.12)$$

Here $C_0 \rho - \omega_0/2 > 0$, and

$$P''(\rho) = 2C_0^2(1 - 3\rho) + 2C_0\omega_0 + 4\pi C_* > 0.$$

At ρ_* the squared term vanishes and the other term is negative, whereas $P(\omega_0) = (1 - \omega_0)\omega_0^2 C_*^2 > 0$. Strict convexity therefore gives a unique switch ρ_{HK} . Here is a fully rational certificate for its locator. Put

$$S_n(x) = \sum_{j=0}^{n-1} \frac{(-1)^j x^{2j+1}}{2j+1}.$$

The alternating series in Machin's identity gives

$$\begin{aligned} \frac{15707963}{5000000} &< 16S_6(1/5) - 4S_3(1/239) < \pi \\ &< 16S_5(1/5) - 4S_2(1/239) < \frac{31415927}{10000000}. \end{aligned}$$

The two outer gaps are respectively

$$\frac{88855616195735287}{1688778771196584375000000}, \quad \frac{3783145391}{215017724250000000}.$$

Set these outer bounds equal to π^- and π^+ , and set

$$\begin{aligned} s_2^- &= \frac{2828427}{2000000}, & s_2^+ &= \frac{1767767}{1250000}, \\ s_3^- &= \frac{4330127}{2500000}, & s_3^+ &= \frac{17320509}{10000000}. \end{aligned}$$

Their square-test residuals are

$$\begin{aligned} 2(2000000)^2 - 2828427^2 &= 705671, & 1767767^2 - 2(1250000)^2 &= 166289, \\ 3(2500000)^2 - 4330127^2 &= 163871, & 17320509^2 - 3(10000000)^2 &= 32019081, \end{aligned}$$

so $s_j^- < \sqrt{j} < s_j^+$. Define

$$\omega^- = \frac{s_3^-}{2\pi^+} - \frac{1}{6}, \quad \omega^+ = \frac{s_3^+}{2\pi^-} - \frac{1}{6}, \quad C_*^\pm = \frac{1}{2} + \frac{8}{15}s_2^\pm, \quad C_0^\pm = 1 + \frac{8}{15}s_2^\pm.$$

For $r_- = 9407/100000$, monotonic substitution gives

$$P(r_-) \leq \mathcal{P}_- := (1 - r_-)(C_0^+ r_- - \omega^-/2)^2 - 2\pi^- C_*^- r_-(\omega^- - r_-),$$

and exact reduction gives

$$\mathcal{P}_- + \frac{1}{5000000} = -\frac{14699034939621959968377942867507773}{2409571458177072753906250000000000000000000} < 0.$$

For $r_+ = 94071/1000000$, one first checks

$$C_0^- r_+ - \frac{\omega^+}{2} - \frac{1}{10} = \frac{619967074526802271}{58904861250000000000} > 0.$$

Thus the square has the same monotonic direction, and

$$P(r_+) \geq \mathcal{P}_+ := (1 - r_+)(C_0^- r_+ - \omega^+/2)^2 - 2\pi^+ C_*^+ r_+(\omega^+ - r_+),$$

where

[illegible]

Consequently

$$\frac{9407}{100000} < \rho_{HK} < \frac{94071}{1000000},$$

such that H_0 is active below it and K_0 above it. The locator is not used as a replacement for the implicit root.

The function K_0 is increasing: a_ρ increases, η_ρ is constant up to $1/2$ and then decreases because $G'_1(s) = -\arccos(s)/\pi$, and the positive root of $\eta y^2 - \sqrt{a}y - C_0 = 0$ increases with a and decreases with η . At $\rho = 5/6$,

$$\cos \frac{9}{10} < 1 - \frac{(9/10)^2}{2} + \frac{(9/10)^4}{24} = \frac{49787}{80000} < \frac{5}{6},$$

so $\arccos(5/6) < 9/10$. Since

$$\eta_{5/6} = \frac{1}{\pi} \int_{5/6}^1 \arccos t \, dt < \frac{1}{6\pi} \arccos \frac{5}{6} < \frac{1}{20},$$

where $\pi > 3$, the positive-root equation implies

$$K_0(5/6) > \frac{a_{5/6}}{\eta_{5/6}^2} > 10 \cdot 3 \cdot 20^2 = 12000, \quad \frac{54}{(1-\rho)^2} \leq 3456 \quad (5/6 \leq \rho < 7/8).$$

Consequently

$$U(\rho) = \begin{cases} H_0(\rho), & \rho_* < \rho < \rho_{HK}, \\ K_0(\rho), & \rho_{HK} \leq \rho < 5/6, \\ 54/(1-\rho)^2, & 5/6 \leq \rho < 7/8. \end{cases} \quad (4.13)$$

The values agree at ρ_{HK} ; the seam face $\rho = 5/6$ is owned by the third branch. For the opposite estimate, the inequality $\arccos t \geq \sqrt{2(1-t)}$, obtained from $\cos u \geq 1 - u^2/2$, gives

$$\eta_{5/6} \geq \frac{1}{\pi} \int_{5/6}^1 \sqrt{2(1-t)} \, dt = \frac{1}{9\pi\sqrt{3}} > \frac{1}{49}.$$

Indeed, $\pi < 22/7$, while $343^2 - 3 \cdot 198^2 = 37 > 0$, so $9\pi\sqrt{3} < 9(22/7)(343/198) = 49$. Also

$$a_{5/6} = 10\pi < \frac{220}{7} < 36, \quad C_0 < 1 + \frac{8}{15} \frac{99}{70} = \frac{307}{175},$$

where $99^2 - 2 \cdot 70^2 = 1 > 0$. Therefore, at $Y = 295$,

$$\eta_{5/6} Y^2 - \sqrt{a_{5/6}} Y - C_0 > \frac{295^2}{49} - 6 \cdot 295 - \frac{307}{175} = \frac{5226}{1225} > 0,$$

so $K_0(5/6) < 295^2$.

For the remaining H_0 check, $\sqrt{3} > 12/7$ and $\pi < 22/7$ give $15\sqrt{3} > 8\pi$, hence $\omega_0 > 1/10$. Also $\sqrt{2} < 3/2$ gives $C_* < 13/10$. Consequently

$$\omega_0 - \frac{1}{16} > \frac{3}{80} > \frac{19}{528}, \quad H_0(1/16) < \frac{13/10}{19/528} = \frac{3432}{95} < 64.$$

Together with the seam ceiling 3456, this proves that $K = 295^2$ is covered at every ratio, including equality.

4.2 Variational comparisons and the complete zero registry

The transformed channel form is

$$\mathfrak{a}_\ell[u] = \int_\rho^1 \left(|u'|^2 + \frac{\ell(\ell+1)}{r^2} |u|^2 \right) dr, \quad D(\mathfrak{a}_\ell) = H_0^1(\rho, 1).$$

Since $r^{-2} \geq 1$, the min-max principle gives, at every radial index,

$$q_{\ell,n}^2 \geq n^2 z_\rho^2 + \ell(\ell+1), \quad q_{0,n} = n z_\rho. \quad (4.14)$$

Extension by zero from $(\rho, 1)$ to $(0, 1)$ embeds the shell fixed-channel form domain isometrically into the ball fixed-channel form domain. The zero trace at $r = \rho$ preserves membership in $H_0^1(0, 1)$, and Hardy's inequality controls the centrifugal term. Thus, preserving both ℓ and n ,

$$q_{\ell,n}^2 \geq j_{\ell+1/2,n}^2. \quad (4.15)$$

The ball forms have the same domain for every angular order; hence, for $p \geq \ell$,

$$j_{p+1/2,n}^2 \geq j_{\ell+1/2,n}^2 + p(p+1) - \ell(\ell+1). \quad (4.16)$$

These are fixed-channel comparisons, not an invocation of unlabelled global domain monotonicity.

We use the exact elementary enclosure

$$\frac{333}{106} < \pi < \frac{22}{7}. \quad (4.17)$$

For the lower bound, Machin's identity $\pi = 16 \arctan(1/5) - 4 \arctan(1/239)$ and alternating sums give a strictly stronger rational bound; for the upper bound,

$$0 < \int_0^1 \frac{x^4(1-x)^4}{1+x^2} dx = \frac{22}{7} - \pi.$$

For clarity we now derive every specialized Bessel-zero inequality used below, apart from the one published first-zero estimate that is stated explicitly before specialization. The spherical Bessel identity and recurrences are standard [5]. Set

$$j_\ell(x) = \sqrt{\frac{\pi}{2x}} J_{\ell+1/2}(x), \quad P_\ell(x) = x^{\ell+1} j_\ell(x).$$

Then

$$P_{\ell+1} = (2\ell+1)P_\ell - x^2 P_{\ell-1}, \quad P'_\ell = x P_{\ell-1}, \quad (4.18)$$

with $P_0 = \sin x$ and $P_1 = \sin x - x \cos x$. In particular,

$$\begin{aligned} P_2 &= (3 - x^2) \sin x - 3x \cos x, \\ P_3 &= (15 - 6x^2) \sin x + (x^3 - 15x) \cos x, \\ P_5 &= (945 - 420x^2 + 15x^4) \sin x + (-945x + 105x^3 - x^5) \cos x, \\ P_6 &= (10395 - 4725x^2 + 210x^4 - x^6) \sin x \\ &\quad + (-10395x + 1260x^3 - 21x^5) \cos x, \\ P_7 &= (135135 - 62370x^2 + 3150x^4 - 28x^6) \sin x \\ &\quad + (-135135x + 17325x^3 - 378x^5 + x^7) \cos x. \end{aligned}$$

Positive zeros are simple by ODE uniqueness. At a zero of j_ℓ , the derivative recurrence gives $j_{\ell+1} = -j'_\ell$. Combined with (4.16), this gives strict interlacing and makes the following sign/count ledger a complete enumeration, rather than a sampled sign test:

endpoint	half-period	signs $j_0, \dots, j_\ell$	zero counts $N_0, \dots, N_\ell$
77/10	$(2\pi, 5\pi/2)$	$+, -$	2, 1
9	$(5\pi/2, 3\pi)$	$+, +, -$	2, 2, 1
10	$(3\pi, 7\pi/2)$	$-, +, +, -, -, -, +$	3, 2, 2, 1, 1, 1, 0
103/10	$(3\pi, 7\pi/2)$	$-, +, +, -$	3, 2, 2, 1
21/2	$(3\pi, 7\pi/2)$	$-, +$	3, 2
23/2	$(7\pi/2, 4\pi)$	$-, -, +, +, -, -, -, +$	3, 3, 2, 2, 1, 1, 1, 0
61/5	$(7\pi/2, 4\pi)$	$-, -, +$	3, 3, 2
129/10	$(4\pi, 9\pi/2)$	$+, -, -, +, +, -$	4, 3, 3, 2, 2, 1

For example, the endpoint signs follow from (4.17), the displayed polynomials, and $\tan s > s$, $\tan s > s + s^3/3$, $\tan s < s/(1 - s^2/2)$ for $0 < s < 1$. The nonautomatic strict comparisons can be taken as

77/10	$5\pi/2 - x > 3/20 > 10/77$
21/2	$7\pi/2 - x > 49/100 > 2/21$
9	$3\pi - x > 21/50 > 9/26$
61/5	$4\pi - x > 9/25 > 915/3646$
103/10	$7\pi/2 - x > 69/100 > 621540/938227$
129/10	$0 < x - 4\pi < 17/50, \quad \tan(x - 4\pi) < 1700/4711$
10	$4/7 < x - 3\pi < 29/50, \quad \tan(x - 3\pi) < 2900/4159$
23/2	$53/50 < 4\pi - x < \pi/2, \quad \tan(4\pi - x) > 546377/375000.$

All comparisons are between positive rationals and are checked by cross multiplication. The ledger proves

$$\begin{aligned} j_{3/2,2} &> 77/10, & j_{5/2,2} &> 9, & j_{7/2,2} &> 103/10, & j_{11/2,2} &> 129/10, \\ j_{3/2,3} &> 21/2, & j_{5/2,3} &> 61/5, \\ j_{13/2,1} &> 10, & j_{15/2,1} &> 23/2. \end{aligned} \tag{4.19}$$

This includes the requested higher-radial derivations. For additional detail, the second root of P_3 is the unique root in $(3\pi, 7\pi/2)$: the derivative chain $R' = x \sin x$, $P'_2 = xR$, $P'_3 = xP_2$ first excludes every earlier cell, and at $x = 103/10$, writing $y = x - 3\pi$, alternating Taylor bounds give $\tan y < 5/4$, while

$$\frac{x(x^2 - 15)}{6x^2 - 15} - \frac{3}{2} = \frac{5917}{621540} > 0,$$

so $P_3(x) < 0$. Likewise, the third root of P_2 is unique in $(7\pi/2, 4\pi)$; at $x = 61/5$,

$$4\pi - x > \frac{97}{265} > \frac{3x}{x^2 - 3},$$

so $P_2(x) > 0$ before its decreasing zero. At $x = 129/10$, the explicit P_5 coefficients and $\tan(x - 4\pi) < 3/8 < |B|/A$ give $P_5(x) < 0$; angular propagation of $j_{5/2,3} > 61/5$ puts the third P_5 -zero above this point, hence its second zero is also above it. These arguments use no root table.

For the remaining first zeros we use Lorch's strict first-zero inequality [4]

$$j_{\nu,1}^2 > \frac{24(\nu + 1)^2}{1 - 2\nu + \sqrt{(2\nu + 3)(2\nu + 11)}} - 2(\nu^2 - 1), \quad \nu > -1.$$

At $\nu = \ell + 1/2$, its right side is

$$R_\ell = \frac{3(2\ell + 3)\sqrt{(\ell + 2)(\ell + 6)} - 2\ell^2 + \ell + 6}{4}.$$

Exact positive-side squaring gives

ℓ	first-zero wall	integer residual proving $R_\ell > c^2$
2	51/10	$105/4 - (51/10)^2 = 6/25$
3	13/2	$81^2 \cdot 5 - 178^2 = 1121$
4	15/2	$66^2 \cdot 15 - 247^2 = 4331$
5	87/10	198189
8	71/6	$1026^2 \cdot 35 - 6067^2 = 35171$
9	64/5	$1575^2 \cdot 165 - 20059^2 = 6939644$
10	69/5	$3450^2 \cdot 3 - 5911^2 = 767579.$

Together with (4.19), this yields the complete first-mode walls

$$\frac{\ell}{\beta_{\ell,1}} \left| \begin{array}{cccccccccc} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 & 10 \\ 4 & 51/10 & 13/2 & 15/2 & 87/10 & 10 & 23/2 & 71/6 & 64/5 & 69/5 \end{array} \right. \quad (4.20)$$

The $\ell = 1$ wall follows internally from $\tan x = x$: the first positive root lies in $(\pi, 3\pi/2)$, and $\tan 4 < 2 < 4$. Applying (4.16) to the higher-radial seeds gives, for later use,

$$j_{p+1/2,2}^2 > 9409/100 + p(p+1) \quad (p \geq 3), \quad (4.21)$$

$$j_{p+1/2,3}^2 > 3571/25 + p(p+1) \quad (p \geq 2), \quad (4.22)$$

$$j_{p+1/2,2}^2 > 13641/100 + p(p+1) \quad (p \geq 5). \quad (4.23)$$

Equations (4.15)–(4.23) transfer every displayed wall to the shell at the same radial index.

4.3 The first and next angular staircases

Throughout the staircase and optical calculations below we use the abbreviation

$$z := z_\rho = \frac{\pi}{1-\rho}.$$

Lemma 4.2 (First two angular bands). *For $\rho_c \leq \rho \leq 7/8$ and $z_\rho \leq K \leq k_2(\rho)$, one has $N_D(A_{\rho,1}, K^2) < W(\rho, K)$.*

Proof. Since $z_\rho > 3$, (4.14) excludes all $n \geq 2$ and all $\ell \geq 2$ through k_2 . Thus the strict caps are

$$N_D = 0 \quad (K = z_\rho), \quad N_D \leq 1 \quad (z_\rho < K \leq k_1), \quad N_D \leq 4 \quad (k_1 < K \leq k_2).$$

At $K = k_m$, the newly possible channel m is excluded because the count is strict. At the first wall,

$$W(\rho, z_\rho) = \frac{2\pi^2}{9} \frac{1+\rho+\rho^2}{(1-\rho)^2} > \frac{2\pi^2}{9} > 2 > 1.$$

For $F(\rho) = W(\rho, k_1(\rho))$, direct logarithmic differentiation gives

$$\frac{F'}{F} = \frac{1+2\rho}{1+\rho+\rho^2} + \frac{2(\pi^2 - (1-\rho)^2)}{(1-\rho)(\pi^2 + 2(1-\rho)^2)} > 0.$$

At ρ_c , exact substitution of $157/50 < \pi < 22/7$ gives

$$F(\rho_c) > \frac{388430104857}{92514296875} = 4 + \frac{18372917357}{92514296875} > 4.$$

Strict increase in K pays both caps and proves the lemma. $\square$

The closed band of Lemma 4.2 meets $\mathcal{D}_{16}$ exactly in $\rho_c \leq \rho < 7/8$, $z_\rho < K \leq k_2$. Indeed $k_2 < K_0$ follows from $K_0 > 64$ and $k_2 < 27$, and on the seam $k_2 < 27 < 1944$. Thus exact subtraction gives

$$\begin{aligned} \mathcal{D}_{17} = & \{\rho_* < \rho < \rho_c : (2\rho)^{-1} < K < U(\rho)\} \\ & \cup \{\rho_c \leq \rho < 7/8 : k_2(\rho) < K < U(\rho)\}. \end{aligned} \quad (4.24)$$

The two former pilot rectangles

$$\begin{aligned} B_0 &= [999/2000, 1001/2000] \times [67/10, 168/25], \\ B_1 &= [999/2000, 1001/2000] \times [168/25, 673/100] \end{aligned}$$

are analytically subsumed. Monotonicity of z_ρ, k_2 and (4.17) give

$$\max z_\rho < \frac{44000}{6993} < \frac{67}{10},$$

$$\min k_2^2 - (168/25)^2 > \frac{126027526}{626250625} > 0, \quad \min k_2^2 - (673/100)^2 > \frac{668749071}{10020010000} > 0.$$

Hence no pilot box is subtracted a second time.

Lemma 4.3 (Staircase through k_5). *For $\rho_c \leq \rho \leq 7/8$ and $k_2(\rho) < K \leq k_5(\rho)$, one has $N_D < W$.*

Proof. Put

$$c_2 = 51/10, \quad c_3 = 13/2, \quad c_4 = 15/2, \quad q_m(\rho) = \max\{k_m(\rho), c_m\}.$$

The first-zero walls in (4.20) and (4.14) give the successive caps

$$4, \quad 9, \quad 16, \quad 25$$

on $K \leq q_2, q_2 < K \leq q_3, q_3 < K \leq q_4$, and $q_4 < K \leq k_5$. Since $z \geq \pi + 1/2 > 7/2$, $4z^2 > z^2 + 30$, so no second radial mode is present. The cumulative multiplicities are $1, 1+3, 1+3+5, 1+3+5+7, 1+3+5+7+9$.

For $F_a(\rho) = (1 - \rho^3)(z_\rho^2 + a)^{3/2}$, the sign of F'_a is the sign of

$$z_\rho^2(1 + \rho) - a\rho^2.$$

It is positive for $a = 6, 12, 20$ on $[\rho_c, 7/8]$. Thus moving payments increase with ρ , while fixed-frequency payments decrease. The following exact split payments cover both one-sided traces:

cap	split	fixed wall on the left	moving wall on the right
9	3/10	$W(3/10, 51/10) > 9$	$W(3/10, k_2) > 9$
16	1/2	$W(1/2, 13/2) = 107653/6336 > 16$	$W(1/2, k_3) > 196/11 > 16$
25	1/2	$W(1/2, 15/2) = 165375/6336 > 25$	$W(1/2, k_4) > 101528/3960 > 25$.

For cap 9, the exact fixed and moving cross-product reserves are 12485961 and 592618642, respectively. Cap 4 is paid by

$$W(\rho_c, k_2) > \frac{7}{99} \frac{342}{343} 73 > 4.$$

All inequalities are strict after a channel enters; equality at a zero wall retains the lower cap.

It remains to compare with the upper owner. If $\rho \leq 1/2$, then $\eta_\rho = \omega_0 < 1/9$: indeed $9\sqrt{3} < 5\pi$ follows after positive squaring from $\pi > 333/106$ and $25 \cdot 333^2 - 243 \cdot 106^2 = 41877 > 0$. Since k_5 increases with ρ ,

$$k_5(\rho)^2 \leq k_5(1/2)^2 = 4\pi^2 + 30 < 4 \left(\frac{22}{7} \right)^2 + 30 = \frac{3406}{49} < 81,$$

and therefore $\eta_\rho k_5(\rho) < 1$.

If $\rho \geq 1/2$, write $\rho = \cos \theta$. Then $\pi \eta_\rho = \sin \theta - \theta \cos \theta \leq \theta(1 - \rho)$, while $\theta \leq \pi/3$ and $\sqrt{z^2 + 30} < z + 15/z$. Hence

$$\eta_\rho k_5(\rho) < \frac{\pi}{3} + \frac{5(1 - \rho)^2}{\pi} \leq \frac{\pi}{3} + \frac{5}{4\pi} < \frac{41}{28} < \frac{131}{75} < C_0.$$

The last chain uses $3 < \pi < 22/7$ and $\sqrt{2} > 7/5$, the latter because $2 > 49/25$. Thus in both ratio cases $\eta_\rho k_5 < C_0$. But $\eta_\rho K_0 = C_0 + \sqrt{a_\rho K_0} > C_0$, hence $k_5 < K_0$. On the seam, with $t = 1 - \rho \leq 1/6$,

$$t^4 k_5^2 = t^2 \pi^2 + 30t^4 < 65/216 < 2916 = t^4 U^2.$$

This proves the lemma on all faces. $\square$

Consequently

$$\begin{aligned} \mathcal{D}_{18} = & \{\rho_* < \rho < \rho_c : (2\rho)^{-1} < K < U(\rho)\} \\ & \cup \{\rho_c \leq \rho < 7/8 : k_5(\rho) < K < U(\rho)\}. \end{aligned} \quad (4.25)$$

4.4 The two-sided staircase

Put

$$\rho_0 = \frac{1}{\sqrt{337}}, \quad d = \frac{\sqrt{337}}{2}, \quad p_1(\rho) = \sqrt{4z_\rho^2 + 2}.$$

Lemma 4.4 (High side through k_6). *For $\rho_c \leq \rho \leq 7/8$ and $z_\rho \leq K \leq k_6(\rho)$, one has $N_D < W$.*

Proof. For $\rho \leq 1/4$, the only possible new modes beyond the first modes $\ell = 0, \dots, 4$ are $(0, 2), (1, 2)$; the first $\ell = 5$ mode is delayed by $j_{11/2,1} > 17/2$. For $\rho \geq 1/4$, $z > 4$ makes every second mode absent through k_6 , and first modes $\ell = 0, \dots, 5$ are the complete inventory. The cumulative first-mode caps are 1, 4, 9, 16, 25, 36; the two possible second modes give caps 26, 29. Their entry walls are

$(1, 1)$	k_1
$(2, 1)$	$\max\{k_2, 51/10\}$
$(3, 1)$	$\max\{k_3, 13/2\}$
$(4, 1)$	$\max\{k_4, 15/2\}$
$(5, 1)$	$\max\{k_5, 17/2\}$
$(0, 2)$	$2z$
$(1, 2)$	$\max\{p_1, 77/10\}$.

For a moving wall $T^2 = az^2 + b$, $W(\rho, T)$ increases whenever $a\pi^2(1 + \rho) > b\rho^2(1 - \rho)^2$, which holds here because $b/a \leq 42$. Fixed walls decrease with ρ . Exact payments are obtained by the splits

$$(k_1, 1/8, 4), (k_2, 3/10, 9), (k_3, 1/2, 16), (k_4, 1/2, 25), (k_5, 13/25, 36), (p_1, 1/5, 29).$$

Writing $A = a(333/106)^2/(1 - r)^2 + b$ and $B = 7(1 - r^3)/99$, each moving payment is the positive rational check $B^2A^3 - C^2 > 0$. In the displayed order the numerators are

$$\begin{aligned} & 153864824754340538785, 1399810848073457853053, 137027505353736631, \\ & 2507119282010251109, 92672404686738742434741, 373255772037478993002833, \end{aligned}$$

over positive denominators. The corresponding fixed-side margins are positive as well; for example they are 1387329/11000000, 6277/6336, 775/704, 452843/343750, and 849901/281250 at the nontrivial splits. Finally $W(\rho_c, 2z) > 26$. Every possible mode is therefore paid, including coincident entries, and every equality remains assigned to the lower cap. Supplement S1 maps every printed moving numerator and fixed margin to its wall and cap. $\square$

Lemma 4.5 (Lower side through d). *For $\rho_0 < \rho < \rho_c$ and $(2\rho)^{-1} < K \leq d$, one has $N_D < W$.*

Proof. Because $3z > d$, every $n \geq 3$ mode is absent. The zero registry and angular propaga-

tion exclude first modes $\ell \geq 6$ and second modes $\ell \geq 3$. The complete cap table is

open-lower, closed-upper range	N_D cap
$L < K \leq 4$	1
$4 < K \leq 51/10$	4
$51/10 < K \leq 13/2, K \leq 2z$	9
$51/10 < K \leq 13/2, K > 2z$	10
$13/2 < K \leq 15/2, K \leq 2z$	16
$13/2 < K \leq 15/2, K > 2z$	17
$15/2 < K \leq 77/10$	26
$77/10 < K \leq 17/2$	29
$17/2 < K \leq 9$	40
$9 < K \leq d$	45.

In fact $13/2 < 2z < 15/2$, so the cap-10 row is empty. The fixed-wall lower bounds, using $\rho < \rho_c < 7/50$, exceed their caps by

$$\frac{793292}{1546875}, \frac{486236661}{1375000000}, \frac{333100003}{990000000}, \frac{329797}{88000}, \frac{3590476297}{1125000000}, \frac{327078887}{990000000}, \frac{8805519}{1375000},$$

respectively. The initial cap is paid by $W(\rho_c, L(\rho_c)) > 19/6 > 1$, and the moving branch by $W(1/19, 2z) > 508/27 > 17$. This proves every row, including $K = d$. The same exact-ledger supplement records each margin with its corresponding row and verifies the moving $2z$ payment. $\square$

The upper-floor comparisons are strict. On the lower side, $H_0(\rho_*) = (2\rho_*)^{-1} > d$, while $K_0 > C_0/\omega_0 > 12 > d$. On the high side, $K_0 > 12 > k_6$ for $\rho \leq 1/2$, $K_0 > 384 > k_6$ for $1/2 < \rho < 5/6$, and the seam owner exceeds $54 > k_6$. Thus exact subtraction gives

$$\begin{aligned} \mathcal{D}_{19} = & \{\rho_* < \rho \leq \rho_0 : (2\rho)^{-1} < K < U(\rho)\} \\ & \cup \{\rho_0 < \rho < \rho_c : d < K < U(\rho)\} \\ & \cup \{\rho_c \leq \rho < 7/8 : k_6(\rho) < K < U(\rho)\}. \end{aligned} \tag{4.26}$$

At $\rho = \rho_0$, $(2\rho_0)^{-1} = d$, so the would-be new lower fiber is empty and the first component owns the equality slice.

4.5 Combined closure of the lower components and the high staircase

Set

$$\sigma := \frac{3\omega_0}{4}.$$

We make every comparison in the required ordering explicit. The alternating Machin sums give

$$\begin{aligned} \pi & < 16 \left(\frac{1}{5} - \frac{1}{3 \cdot 5^3} + \frac{1}{5 \cdot 5^5} - \frac{1}{7 \cdot 5^7} + \frac{1}{9 \cdot 5^9} \right) \\ & \quad - 4 \left(\frac{1}{239} - \frac{1}{3 \cdot 239^3} \right) < \frac{355}{113}, \end{aligned}$$

and the last gap is exactly

$$\frac{45167474711}{189820334689453125} > 0.$$

Since

$$3 \cdot 1101^2 \cdot 113^2 - 607^2 \cdot 355^2 = 1998482 > 0,$$

we have

$$\sqrt{3} > \frac{607 \cdot 355}{1101 \cdot 113} > \frac{607\pi}{1101},$$

and therefore

$$\omega_0 = \frac{\sqrt{3}}{2\pi} - \frac{1}{6} > \frac{40}{367}. \quad (4.27)$$

On the other side,

$$25 \cdot 333^2 - 243 \cdot 106^2 = 41877 > 0$$

and $\pi > 333/106$ imply $9\sqrt{3} < 5\pi$, hence $\omega_0 < 1/9$. Moreover $2 \cdot 32^2 - 45^2 = 23 > 0$, so $\sqrt{2} > 45/32$, $C_0 > 7/4$, and

$$\rho_* = \frac{\omega_0}{2C_0} < \frac{2}{63}.$$

Now all remaining comparisons are witnessed by

$$\begin{aligned} 4 \cdot 337 < 63^2, \quad 18^2 < 337, \quad \frac{1}{18} < \frac{3}{40} < \frac{3\omega_0}{4} < \frac{1}{12} < \frac{7}{51}, \\ \frac{7}{51} < \frac{1}{1+2\pi} < \frac{1}{7}, \quad \frac{1}{7} < \frac{39}{50} < \frac{7}{8}. \end{aligned}$$

Here the first inequality on the second line uses $\pi < 22/7$, and the second uses $\pi > 3$. Thus, without decimal comparison,

$$\rho_* < \rho_0 < \sigma < \rho_c < \frac{39}{50} < \frac{7}{8}.$$

Lemma 4.6 (Closure of both lower components of $\mathcal{D}_{19}$). *The strict inequality $N_D < W$ holds on the first two components of (4.26).*

Proof. We divide the proof into a wedge, a finite staircase, and one remaining floor cell.

For a low-interface tail beginning at $x_r = r + 1/2 < \mu = \rho K$, put

$$n_r = \lfloor \mu - x_r \rfloor, \quad q_r = x_r + n_r, \quad p = \lfloor \omega_0 K \rfloor.$$

The concave-head/convex-tail split used in (4.6), now with its coarse first-shelf payment, gives

$$\frac{\mathcal{T}_r}{2} < \int_{x_r}^K A_{\rho,K}(z) dz + \delta_r - \frac{p - n_r}{4}, \quad 0 \leq \delta_r < \frac{2(\mu - q_r)^{5/2}}{15\sqrt{\mu}}. \quad (4.28)$$

When $n_r = 0$, the concave head is absent and the formula follows from the convex tail alone. From (4.27) and $400 \cdot 337 - 367^2 = 111 > 0$,

$$\omega_0 d > \frac{40}{367} \frac{\sqrt{337}}{2} = \frac{20\sqrt{337}}{367} > 1.$$

If $0 < \rho \leq \sigma$, $K > d$, and $\rho K > 1/2$, then with $x = \omega_0 K > 1$ and $m = \lfloor x \rfloor$,

$$n_r \leq \left\lfloor \frac{3x}{4} - \frac{1}{2} \right\rfloor < m = p.$$

Since $\delta_r < 2\sqrt{2}/15 < 1/4$, every tail in (4.28) is strict. The hypotheses hold throughout the first component of $\mathcal{D}_{19}$ and on $\rho_0 \leq \rho \leq \sigma$, $K > d$. This includes the complete $\rho = \rho_0$ fiber.

For $\rho_0 < \rho < \rho_c$ and $K \leq \sqrt{114}$, the complete zero registry and (4.16) give the following exhaustive inventory:

range (lower face open, upper face closed)	newly possible modes	cap
$d < K \leq 10, K \leq 3z$	first 0:5, second 0:2	45
$d < K \leq 10, K > 3z$	preceding plus (0, 3)	46
$10 < K \leq 81/8$	plus (6, 1), possibly (0, 3)	59
$81/8 < K \leq 21/2$	plus (3, 2)	66
$21/2 < K \leq \sqrt{7073}/8$	plus (1, 3)	69
$\sqrt{7073}/8 < K \leq \sqrt{114}$	plus (4, 2)	78.

First modes $\ell \geq 7$, second modes $\ell \geq 5$, third modes $\ell \geq 2$, and all $n \geq 4$ are absent. In particular,

$$(81/8)^2 + 8 = 7073/64$$

is the propagated (4, 2) wall. Since $\rho < \rho_c < 1/7$, the uniform estimate

$$W(\rho, K) > \frac{2}{9(22/7)} \left(1 - \frac{1}{7^3}\right) K^3$$

pays the successive caps with exact positive reserves

$$\frac{2908}{539}, \quad \frac{6199}{539}, \quad \frac{990435}{137984}, \quad \frac{555}{44}, \quad \frac{159}{44}.$$

The first number pays even cap 46 from $K > 9$; the last pays cap 78 from $K > 21/2$. Hence $K = \sqrt{114}$ is included.

It remains to consider $\sigma < \rho < \rho_c$ and $K > \sqrt{114}$. Put

$$p = \lfloor \omega_0 K \rfloor, \quad m = \lfloor \rho K - 1/2 \rfloor.$$

Then $p \geq 1$. Except on

$$\mathcal{B} = \left\{ \sigma < \rho < \rho_c, \quad K > \sqrt{114}, \quad \rho K \geq 5/2, \quad K < 2/\omega_0 \right\}, \quad (4.29)$$

one has $m \leq 2p - 1$. Indeed this is immediate when $\rho K < 5/2$; when $K \geq 2/\omega_0$, one has $p \geq 2$. Moreover,

$$\rho_c < \frac{1}{7} < \frac{60}{367} < \frac{3\omega_0}{2},$$

where the first and last comparisons were proved above and $367 < 420$. Hence $\rho K - 1/2 < 2p$.

Away from a half-integer μ , all interface remainders in (4.28) are equal. Summing over $r = 0, \dots, m$ gives the average floor reserve

$$\frac{1}{4} \left(p - \frac{m}{2} \right) \geq \frac{1}{8}.$$

For $m = 0$, use $\delta < 1/4$; for $m \geq 1$, $\mu > 3/2$ and $\delta < 2/(15\sqrt{3/2}) < 1/8$. At $\mu \in \mathbb{Z} + 1/2$, assign the interface to the high side; then the remainder is zero and strictness remains in the nonzero convex tail.

On $\mathcal{B}$, $\omega_0 > 40/367$, so $K < 367/20 =: R$. The turning estimate

$$A_{\rho, K}(z) \leq G_K(z) < \frac{(K - z)^{3/2}}{3\sqrt{K}}$$

and monotonicity in K reduce the proxy count at $z_\ell = \ell + 1/2$ to

$$\frac{(357 - 20\ell)^3}{1321200} < \left(c_\ell + \frac{3}{4} \right)^2.$$

The integer numerators of these strict comparisons are

ℓ	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14
c_ℓ	6	5	5	4	4	3	3	3	2	2	1	1	1	1	0

and, in the same order,

$$14697882, 5409422, 11827162, 3611502, 8555642, 1604782, 5267322, 8361062, \\ 2346202, 4446342, 176282, 1474822, 2444562, 3133502, 286642.$$

The $\ell = 14$ row and monotonicity remove the entire angular tail. Therefore the full multiplicity cap is

$$\sum_{\ell=0}^{13} (2\ell + 1)c_\ell = 395.$$

But $\rho K \geq 5/2$, $\rho < \rho_c < 11/80$, and $1/\pi > 113/355$ give

$$W(\rho, K) > \frac{160293325}{378004} = 395 + \frac{10981745}{378004} > 395.$$

Thus $\mathcal{B}$ is closed as well. The face $\rho K = 5/2$ is $\mathcal{B}$ -owned and $K = 2/\omega_0$ is aggregate-owned. All lower components of $\mathcal{D}_{19}$ are proved. $\square$

We now turn to the high component. Define the strict channel proxy

$$b_{\ell,n}(z)^2 = \max\{n^2 z^2 + \ell(\ell + 1), \beta_{\ell,n}^2\}, \quad (4.30)$$

For the radial channel we set, for every $n \geq 1$,

$$\beta_{0,n} := j_{1/2,n} = n\pi,$$

the last identity following from $J_{1/2}(x) = \sqrt{2/(\pi x)} \sin x$. For $\ell \geq 1$, the first-mode β 's are in (4.20) and

$$\begin{aligned} \beta_{1,2} &= 77/10, & \beta_{2,2} &= 9, & \beta_{3,2} &= 103/10, & \beta_{4,2} &= \sqrt{11409}/10, \\ \beta_{5,2} &= 129/10, & \beta_{1,3} &= 21/2, & \beta_{2,3} &= 61/5. \end{aligned}$$

By (4.14) and (4.15), a mode can be counted only if $b_{\ell,n}(z) < K$.

Lemma 4.7 (Closed high staircase through k_{11}). *For $\rho_c \leq \rho \leq 7/8$ and $z_\rho \leq K \leq k_{11}(\rho)$, one has $N_D < W$.*

Proof. The exhaustive checkpoint inventories are

checkpoint	possible modes
k_7	first 0:6, second 0:1, no $n \geq 3$
k_8	first 0:7, second 0:3, no $n \geq 3$
k_9	first 0:8, second 0:4, no $n \geq 3$
k_{10}	first 0:9, second 0:5, third 0:1, no $n \geq 4$
k_{11}	first 0:10, second 0:4, third 0:1, no $n \geq 4$.

For example, at k_{11} the mode $(5, 2)$ would require both $z^2 < 34$ from the product wall and $z^2 > (129/10)^2 - 132 = 3441/100$ from the ball wall, a contradiction. Also $4z > k_{11}$ at $z \geq \pi + 1/2$. Thus radial indices 1, 2, 3, 4 and angular indices through 11 are proved above to be exhaustive for the stated checkpoints.

Writing H for the highest first channel and h for the highest second channel, the block cap is $(H+1)^2 + (h+1)^2$, with an additional 1 or 4 for third modes. The complete cap tables at k_8 and k_9 are

$k_8 : H \setminus h$	$\emptyset$	0	1	2	3
7	64	65	68	73	80
6	49	50	53	—	—
5	36	37	40	45	52
≤ 4	25	26	29	34	41

$k_9 : H \setminus h$	$\emptyset$	0	1	2	3	4
8	81	—	—	—	—	—
7	64	65	68	73	—	—
6	49	50	53	58	65	74
5	36	37	40	45	52	61
≤ 4	25	26	29	34	41	50

The dashes follow from incompatible necessary inequalities, not from a numerical sample. At k_{10} the cap lists are

$$100; \quad 81, 97; \quad 64, 80, 89, 100; \quad 74, 69, 78, 85, 89,$$

and at k_{11} they are

$$121; \quad 100, 125; \quad 106, 110; \quad 93.$$

Define the exact finite proxy

$$C_z(K) = \sum_{(\ell, n)} (2\ell + 1) \mathbf{1}_{\{b_{\ell, n}(z) < K\}},$$

where the sum runs over the finite inventory above. Between proxy events it is constant and W increases in K . Every squared event is either constant or affine in z^2 . At a moving event $K^2 = sz^2 + d_0$, the derivative of its Weyl payment has the sign of

$$s\pi^2 \frac{1 + \rho}{(1 - \rho)^2} - \rho^2 d_0. \quad (4.31)$$

Thus the finite split registry

$$\frac{1}{5}, \frac{3}{10}, \frac{1}{3}, \frac{3}{8}, \frac{2}{5}, \frac{5}{12}, \frac{3}{7}, \frac{1}{2}, \frac{107}{200}, \frac{8}{15}, \frac{3}{5}, \frac{16}{25}, \frac{13}{20}, \frac{2}{3}, \frac{4}{25}, \frac{1}{4}, \frac{7}{20},$$

together with $z^2 = 16, 68/3, 34$, reduces every cell to exact rational endpoint arithmetic. The smallest representative reserves in the six successive bands are

$$\frac{1387329}{11000000}, \quad \frac{849901}{281250}, \quad \frac{835355}{577368}, \quad \frac{835355}{577368}, \quad \frac{943655}{171072}, \quad \frac{507845}{261954}.$$

Every realizable event is strictly paid, with coincident events charged at their combined multiplicity.

There is one conservative row requiring explicit treatment. Angular propagation gives

$$j_{9/2,2}^2 > \left(\frac{103}{10}\right)^2 + 8 = \frac{11409}{100}.$$

If $(4, 2)$ were counted at k_9 , this ball wall would force $z^2 > 2409/100$, whereas the product wall forces $z^2 < 70/3$; since $2409/100 - 70/3 = 227/300 > 0$, the printed cap-74 cell is actually empty. If one deliberately retains the weaker conservative cell, counting the mode forces $K > 207/20$ and $\rho < 7/20$. Therefore

$$W(\rho, K) > \frac{52823261673}{704000000} = 74 + \frac{727261673}{704000000} > 74. \quad (4.32)$$

This is the corrected cap-74 payment; it is smaller than the generic k_9 summary reserve and must not be hidden by a claim that all omitted cells have larger reserves.

At every $K = k_m$, the $(m, 1)$ mode is on its product equality wall and is excluded. At every fixed Bessel wall the zero inequality is strict. The event calculation therefore covers all equality and coincidence faces, and proves the lemma. $\square$

The combined finite analytic ledger—covering the zero/sign registry, the two-sided and lower staircases, the lower closure, the high staircase, optical constants, and exact subtraction—can be replayed without floating point. Its core contains 488 substantive rational/radical checks and 65 bookkeeping identities; these are whole-ledger counts, not a count of k_{11} -only cells. It also checks the sign of (4.31) on the named faces. Every proof decision is a cross multiplication or positive-side squaring; decimal displays play no role. The complete deterministic registry and executable source are printed in Supplement S1.

4.6 The optical closure and the exact residual $\mathcal{D}_{20}$

Lemma 4.8 (All-frequency optical theorem). *For $39/50 \leq \rho < 1$ and $K \geq 0$, $N_D(A_{\rho,1}, K^2) \leq W(\rho, K)$, with strict inequality for $K > 0$.*

Proof. Put $\varepsilon = 1 - \rho \leq 11/50$, $a = \varepsilon K$, and $c = 1126/625$. We use two inclusive screens meeting at $a = c/\varepsilon$.

For the product screen, (4.14) implies that a counted mode satisfies

$$n^2 \pi^2 + \varepsilon^2 \ell(\ell + 1) < a^2.$$

There is no mode for $a \leq \pi$. For $a > \pi$, write $a/\pi = N + t$, with $N \geq 1$, $0 < t \leq 1$, assigning $(N, t) = (m - 1, 1)$ at $a = m\pi$. If

$$M_n = \left\lceil \sqrt{(b_n/\varepsilon)^2 + 1/4} - 1/2 \right\rceil,$$

the exact angular multiplicity is bounded by

$$\mathcal{P}_\varepsilon(a) = \sum_{n=1}^N M_n^2.$$

Direct summation of the continuous-minus-discrete product deficit gives

$$\frac{D(a)}{\pi^2} = \frac{N^2}{2} + N \left(t^2 + \frac{1}{6} \right) + \frac{2t^3}{3}. \quad (4.33)$$

For $C_D = 1382/3125$, the difference $D/\pi^2 - C_D(N + t)^2$ increases in N , and at $N = 1$ is minimized at $t = C_D$, where it equals $1822532/91552734375 > 0$. Thus $D > C_D a^2$.

The elementary ceiling bound and quarter-circle integral give

$$\varepsilon^2 \mathcal{P}_\varepsilon < S_0 + \varepsilon a^2/4 + \varepsilon^2 a/\pi.$$

On $a \leq c/\varepsilon$, using $q = 106/333 > 1/\pi$ and $\varepsilon \leq 11/50$, the exact reserve is

$$C_D - \left(\frac{2cq}{3} + \frac{11}{200} + (11/50)^2 q^2 \right) = \frac{39569}{2772225000} > 0. \quad (4.34)$$

This proves the low screen, including all radial and angular equality walls.

For the complementary-action screen, use the inverse-action construction proved in Lemma 3.4: let the decreasing shell action have inverse $R_{\mathcal{A}}$, put $F = R_{\mathcal{A}}^2$, and $g = -F'$. If τ is the actual curvature interface and $T = a/\pi$, that lemma gives

$$g \text{ decreasing on } (0, \tau), \quad g \text{ increasing on } (\tau, T),$$

$$F(0) = a^2, \quad F(\tau) = \rho^2 a^2, \quad F(T) = 0, \quad g(T) = 2\pi \rho a.$$

For $C(t) = \#\{n : n - 1/4 < t\}$, set $w = t - C(t)$ and $\Psi(t) = \int_0^t w(s) ds - t/4$. The complete sawtooth and Stieltjes calculation in Lemma 3.6, including the ungridded curvature interface τ , gives

$$D_{\text{rad}} := \int_0^T F(t) dt - \sum_{n-1/4 < T} F(n - 1/4) \geq \frac{\rho^2 a^2}{4} - \frac{\pi \rho a}{4}. \quad (4.35)$$

The exact half-integer angular count from Lemma 3.5 satisfies $\varepsilon^2 M_\varepsilon(x)^2 < (x + \varepsilon/2)^2$, even on an angular wall. Since fewer than $a/\pi + 1/4$ radial layers occur, the angular error is strictly below

$$E = \left(\frac{a}{\pi} + \frac{1}{4} \right) \left(\varepsilon a + \frac{\varepsilon^2}{4} \right). \quad (4.36)$$

On $a \geq c/\varepsilon$, the lower screen from (4.35) decreases and the upper screen from (4.36) increases with ε . At $\varepsilon = 11/50$, their exact difference is

$$\frac{14817541}{472867032960000} > 0. \quad (4.37)$$

Thus the strict layer-cake sum is below the action integral, and the phase bridge proves the high screen. Both screens include $a = c/\varepsilon$; at $K = 0$ both sides vanish. $\square$

Combining Lemmas 4.6, 4.7, and 4.8 with (4.26) leaves exactly

$$\mathcal{D}_{20} = \left\{ (\rho, K) : \rho_c \leq \rho < \frac{39}{50}, \quad k_{11}(\rho) < K < U(\rho) = K_0(\rho) \right\}. \quad (4.38)$$

Indeed $k_{11} < U$: on $\rho \geq \rho_c$, $a_\rho \geq 1$ and $\eta_\rho < 1/8$, so $K_0 > 64$, whereas $k_{11} < 64$ for $\rho < 7/8$; the seam owner is greater than 1944. On the surviving ratio range neither H_0 nor the seam branch is eligible, so $U = K_0$. The face $K = k_{11}$ belongs to the closed staircase, $\rho = 39/50$ belongs to the optical theorem, and $K = U$ belongs to the fixed-ratio high-frequency owner.

4.7 Certified closure of the exact residual $\mathcal{D}_{20}$

The last step has two parts: a finite compact rectangle and an analytic tail in K , whose three base signs are certified at $K = 200$. We state the certificate predicates explicitly so that the machine's role is separated from the analytic proof.

Lemma 4.9 (Compact rectangle). *On the complete closed rectangle*

$$\frac{7}{51} \leq \rho \leq \frac{39}{50}, \quad 12 \leq K \leq 200,$$

one has $N_D(A_{\rho,1}, K^2) < W(\rho, K)$.

Proof. For fixed ν , direct differentiation, with the zero extension at $a = \nu$, gives

$$\partial_a G_a(\nu) = \frac{1}{\pi} \sqrt{1 - \frac{\nu^2}{a^2}} \quad (a > \nu).$$

Thus $A_{\rho,K}(\nu)$ decreases weakly with ρ . For fixed ρ ,

$$\partial_K A_{\rho,K}(\nu) = \begin{cases} \sqrt{K^2 - \nu^2}/(\pi K), & \rho K \leq \nu < K, \\ (\sqrt{K^2 - \nu^2} - \sqrt{\rho^2 K^2 - \nu^2})/(\pi K), & \nu < \rho K, \end{cases}$$

and both branches are positive and meet continuously. The newly active term at a half-integer $K = \nu_\ell$ has strict count $[1/4]_< = 0$. Consequently P_{coarse} decreases weakly with ρ and increases weakly with K , across both interfaces. Since

$$W_\rho = -\frac{2\rho^2 K^3}{3\pi} < 0, \quad W_K = \frac{2(1 - \rho^3)K^2}{3\pi} > 0,$$

every closed rational box $B = [\rho_-, \rho_+] \times [K_-, K_+]$ satisfies

$$P_{\text{coarse}}(\rho, K) \leq P_{\text{coarse}}(\rho_-, K_+), \quad W(\rho, K) \geq W(\rho_+, K_-). \quad (4.39)$$

At the proxy corner, for each exact active half-integer $\nu_\ell < K_+$, outward ball arithmetic encloses

$$X_\ell = A_{\rho_-, K_+}(\nu_\ell) + 1/4.$$

It proposes an integer M_ℓ from the outward upper endpoint but accepts it only after the certain comparison $X_\ell \leq M_\ell$. Then

$$[X_\ell]_< \leq M_\ell - 1, \quad \hat{P}_B := \sum_{\nu_\ell < K_+} (2\ell + 1)(M_\ell - 1). \quad (4.40)$$

This is correct even at $X_\ell = M_\ell$; an ordinary floor would be wrong there. At the opposite corner, an outward lower ball $W_{B,-} \leq W(\rho_+, K_-)$ is computed. A leaf is accepted only if

$$W_{B,-} - \hat{P}_B > 0 \quad (4.41)$$

is a certain ball comparison. Equations (4.1), (4.39), (4.40), and (4.41) then prove the strict theorem on the entire closed leaf.

Here is the complete deterministic partition algorithm.

1. Start with the exact rational root $[7/51, 39/50] \times [12, 200]$, a stack containing that root at depth zero, and empty caches for the two monotone corners.
2. Evaluate (4.41) at 256-bit Arb precision. If it passes, store the four rational endpoints, depth, and integer $\hat{P}_B$.
3. If it fails, bisect either ρ or K at the exact rational midpoint. Both possible bisections are evaluated; a display-only score chooses the direction, but this score never accepts a box. Push the two chosen children, lower child first. Stop with failure if depth would exceed 30 or the box count would exceed 100000.
4. For G_a , decide $\nu < a$, $\nu = a$, or $\nu > a$ by exact rational comparison before calling square root or arccos. Enumerate the active set as $\max\{0, \lceil K - 1/2 \rceil\}$, so $\nu = K$ is excluded.
5. After termination, sort leaves canonically. Check positive rational width, exact total area, every elementary open ρ -slab, and every vertical rational face. Point ownership is lower-closed and upper-open in each coordinate, except that the two outer upper faces are closed. The analytic inequality itself was proved on every closed leaf.

The run terminates with 10580 leaves, maximum depth 15, 16020 distinct proxy corners, and 2153076 certified Arb-versus-integer comparisons. There are zero unresolved integer-wall straddles. Exact slab, face, and area sweeps certify full coverage and unique half-open ownership. At 256 bits the canonical digest of the schema, precision, root, and every sorted leaf record is

c0f92eeda050934a147ab6e774c3968c49c38517fc63e46d50d3b4c617fb1a72.

A separate 384-bit regeneration produces the same leaf count and all strict signs; its precision-dependent digest is

4ee16f3bd7e7659fb38b4f63fd51ce0d36740581f7786eb502469da5323fb7aa.

The complete executable that regenerates both values is printed in Supplement S2, *Executable interval certificates*. Thus the finite predicates just described prove the lemma. $\square$

We next reconstruct the aggregate theorem, including the discrete floor arithmetic that links its continuous reserve to the spectrum. Put

$$d_\rho = \frac{2 \arcsin \rho}{\pi}, \quad \mu = \rho K, \quad C = \frac{2\pi\rho K}{1-\rho}.$$

When $\mu > 1/2$, write uniquely

$$\mu - \frac{1}{2} = J + \tau, \quad J \in \mathbb{N}_0, \quad 0 \leq \tau < 1,$$

and define

$$R = \begin{cases} J, & \tau = 0, \\ J + 1, & \tau > 0, \end{cases} \quad x_r = r + \frac{1}{2} \quad (0 \leq r < R).$$

Then

$$n_r = \lfloor \mu - x_r \rfloor = J - r, \quad q_r = x_r + n_r = J + \frac{1}{2}, \quad \sum_{r=0}^{R-1} n_r = \frac{J(J+1)}{2}, \quad \mu - q_r = \tau. \quad (4.42)$$

For $C > 0$, set

$$\mathcal{I}(C, R) = \frac{1}{2} \left[R\sqrt{R^2 + C} + C \operatorname{arsinh} \frac{R}{\sqrt{C}} - R^2 \right]. \quad (4.43)$$

Lemma 4.10 (Discrete aggregate implication). *Define*

$$\mathcal{Q}(\rho, K) = R \lfloor K\eta_\rho \rfloor + d_\rho \frac{J(J+1)}{2} - (1 + d_\rho) \mathcal{I}(C, R) - \frac{8R\tau^{5/2}}{15\sqrt{\mu}}. \quad (4.44)$$

If $\mathcal{Q}(\rho, K) \geq 0$, then $N_D(A_{\rho,1}, K^2) < W(\rho, K)$.

Proof. For the accepted first-shelf length p_r , the curvature bound gives

$$p_r < \sqrt{x_r^2 + C} - x_r =: f_C(x_r).$$

The function f_C is positive and strictly convex. The midpoint inequality therefore yields the strict sum

$$\sum_{r=0}^{R-1} p_r < \sum_{r=0}^{R-1} f_C(r + 1/2) \leq \int_0^R f_C(x) dx = \mathcal{I}(C, R). \quad (4.45)$$

The interface remainder common to all low tails is zero at $\tau = 0$ and otherwise satisfies

$$0 \leq \delta_\tau \leq \frac{2\tau^{5/2}}{15\sqrt{\mu}}.$$

The exact concave-head/convex-tail estimate is

$$\frac{\mathcal{T}_r}{2} \leq \int_{x_r}^K A(z) dz + \delta_\tau - \frac{1}{4} (\lfloor K\eta_\rho \rfloor + d_\rho n_r - (1 + d_\rho) p_r).$$

This formula also holds when $n_r = 0$, with absent head and $p_r = 0$. Summing and applying (4.42) and (4.45) gives

$$\sum_{r < R} \frac{\mathcal{T}_r}{2} - \sum_{r < R} \int_{x_r}^K A(z) dz < -\frac{1}{4} \mathcal{Q}(\rho, K). \quad (4.46)$$

For $r \geq R$, $x_r \geq \mu$, including equality at a half-integer interface; zero extension and the convex tail inequality give $\mathcal{T}_r \leq 2 \int_{x_r}^K A$. Tails beginning at or above K vanish.

If $h_\ell = [A(\ell + 1/2) + 1/4]_<$, the exact multiplicity identity is

$$\sum_{\ell \geq 0} (2\ell + 1) h_\ell = \sum_{r \geq 0} \left(h_r + 2 \sum_{\ell > r} h_\ell \right).$$

Each parenthesis is bounded by $\mathcal{T}_r$; at an integer proxy wall the strict count is one smaller than the ordinary floor. Hence $\mathcal{Q} \geq 0$ and (4.46) imply

$$P_{\text{coarse}} < 2 \sum_{r \geq 0} \int_{x_r}^K A(z) dz = 2 \int_0^K f_3(z) A(z) dz.$$

The weighted scaffold satisfies $\int_0^z f_3(t) dt \leq z^2/2$. Integration by parts, using that A decreases and $A(K) = 0$, gives

$$2 \int_0^K f_3 A \leq \int_0^K 2z A(z) dz = W(\rho, K).$$

The first inequality is strict, including $\tau = 0$, every half-integer μ , and every integer $K\eta_\rho$. Together with (4.1), this proves the lemma. $\square$

For $K\eta_\rho > 1$, define

$$b = \frac{2\pi\rho}{1-\rho}, \quad \bar{R} = \mu + \frac{1}{2}, \quad S = \sqrt{\bar{R}^2 + bK},$$

$$\bar{\mathcal{I}} = \frac{1}{2} \left[\bar{R}S + bK \operatorname{arsinh} \frac{\bar{R}}{\sqrt{bK}} - \bar{R}^2 \right]$$

and the continuous reserve

$$\begin{aligned} \mathcal{B}(\rho, K) &= (\mu - 1/2)(K\eta_\rho - 1) + \frac{d_\rho}{2}(\mu - 3/2)(\mu - 1/2) \\ &\quad - (1 + d_\rho)\bar{\mathcal{I}} - \frac{8(\mu + 1/2)}{15\sqrt{\mu}}. \end{aligned} \quad (4.47)$$

If $y = K\eta_\rho > 1$ and $\mu > 3/2$, then

$$R[y] > R(y - 1) \geq (\mu - 1/2)(y - 1),$$

$$\frac{d_\rho J(J+1)}{2} > \frac{d_\rho}{2}(\mu - 3/2)(\mu - 1/2), \quad \mathcal{I}(C, R) < \bar{\mathcal{I}}, \quad R\tau^{5/2} < \bar{R}.$$

Therefore

$$\mathcal{Q}(\rho, K) > \mathcal{B}(\rho, K), \quad \mathcal{B} \geq 0 \implies \mathcal{Q} > 0 \implies N_D < W. \quad (4.48)$$

Lemma 4.11 (Aggregate tail for $K \geq 200$). *For $\rho_c \leq \rho \leq 39/50$ and $K \geq 200$, one has $N_D(A_{\rho,1}, K^2) < W(\rho, K)$.*

Proof. It suffices to work on the rational superset $7/51 \leq \rho \leq 39/50$. On it,

$$1 - \rho \geq 11/50, \quad \mu \geq 1400/51 > 3/2.$$

Moreover $G_1(z) = \pi^{-1} \int_z^1 \arccos t \, dt$, and $\arccos t \geq \sqrt{2(1-t)}$, so

$$200\eta_\rho \geq 200G_1(39/50) > \frac{1400\sqrt{2}}{33}(11/50)^{3/2} > 1;$$

the square of the last left side is $1043504/27225 > 1$. Thus all definitions above are valid and $S > \bar{R} > \rho K$. At $\rho = 1/2$, the constant and variable definitions of η_ρ agree exactly.

For fixed ρ , write $I(K) = \bar{I}$ and $P = S - \bar{R}$. Differentiation of

$$I(K) = \int_0^{\bar{R}} (\sqrt{x^2 + bK} - x) \, dx$$

gives

$$I_K = \rho P + \frac{b}{2} \operatorname{arsinh} \frac{\bar{R}}{\sqrt{bK}}, \quad (4.49)$$

$$I_{KK} = \rho^2 \left(\frac{\bar{R}}{S} - 1 \right) + \frac{\rho b}{2S} + \frac{b(2\rho K - 1)}{8KS}. \quad (4.50)$$

Consequently

$$\begin{aligned} \mathcal{B}_K &= \rho(K\eta_\rho - 1) + (\mu - 1/2)\eta_\rho + d_\rho\rho(\mu - 1) \\ &\quad - (1 + d_\rho)I_K - \frac{2\rho(2\mu - 1)}{15\mu^{3/2}}, \end{aligned} \quad (4.51)$$

$$E_{KK} = \frac{\rho^2(2\mu - 3)}{15\mu^{5/2}}, \quad (4.52)$$

$$\mathcal{B}_{KK} = 2\rho\eta_\rho + d_\rho\rho^2 - (1 + d_\rho)I_{KK} + E_{KK}. \quad (4.53)$$

Since $S > \bar{R} > \rho K$, the first term in (4.50) is negative and

$$I_{KK} < \frac{3\rho b}{4S} < \frac{3b}{4K} \leq \frac{3b}{800}, \quad E_{KK} > 0. \quad (4.54)$$

Define

$$F(\rho) = 2\rho\eta_\rho + d_\rho\rho^2 - \frac{3(1 + d_\rho)b}{800}.$$

Equations (4.53)–(4.54) prove, for every $K \geq 200$,

$$\mathcal{B}_{KK}(\rho, K) > F(\rho). \quad (4.55)$$

The finite certificate at the single base frequency $K = 200$ establishes on $[7/51, 39/50]$ the three strict predicates

$$\mathcal{B}(\rho, 200) > 0, \quad \mathcal{B}_K(\rho, 200) > 0, \quad F(\rho) > 0. \quad (4.56)$$

We describe its exact validation below. Assuming only those certified base signs, (4.55) gives $\mathcal{B}_{KK} > 0$ at every frequency, and the two exact integrations

$$\mathcal{B}_K(\rho, K) = \mathcal{B}_K(\rho, 200) + \int_{200}^K \mathcal{B}_{KK}(\rho, s) \, ds > 0,$$

$$\mathcal{B}(\rho, K) = \mathcal{B}(\rho, 200) + \int_{200}^K \mathcal{B}_K(\rho, s) ds > 0$$

prove the conclusion for $K > 200$; the base predicates own $K = 200$. Finally apply (4.48).

For completeness, the base certificate partitions at the formula interface $\rho = 1/2$. Starting from $7/51$, it creates consecutive exact rational intervals of width $1/2000$, shortening only the last interval of each segment. There are 726 intervals on $[7/51, 1/2]$ and 560 on $[1/2, 39/50]$, hence 1286 boxes. Each is converted to an outward Arb ball by its exact rational midpoint and radius; the verifier first proves that both exact endpoints are contained. At 192-bit precision it then requires certain signs for (4.56) and, as consistency checks only,

$$1 - \rho > 0, \quad \mu > 3/2, \quad 200\eta_\rho > 1, \quad S > \bar{R} > 200\rho,$$

$$I_{KK} < 3b/800, \quad E_{KK} > 0, \quad \mathcal{B}_{KK} > F.$$

No base consistency check is extrapolated to $K > 200$; that is the role of (4.54) and the two integrations. Exact endpoint adjacency proves that the boxes tile both segments with no gap or overlap. A 384-bit replay gives the same strict signs. The run performs 14146 certain sign comparisons and 2572 exact-endpoint containment checks at each precision. The canonical certificate digests at 192 and 384 bits are, respectively,

54e115a05a9257b5b9279c7f79bfe4b866388d2dd2868101181da17ab4bdfc50,

6f0bc0fcb26cfa78b12d95fa51b20bf3238d801e61d4b11248253f3a2075df92.

The complete executable is printed in Supplement S2. $\square$

Proposition 4.12 (Exact closure of $\mathcal{D}_{20}$). *The compact-middle residual is empty after the preceding two lemmas.*

Proof. First, $\pi < 22/7$ gives

$$14\pi < 44 \implies 7(1 + 2\pi) < 51 \implies \frac{7}{51} < \rho_c.$$

Second, on the correctly scoped domain $\rho_c \leq \rho < 1$,

$$z_\rho \geq z_{\rho_c} = \pi + \frac{1}{2} > \frac{7}{2},$$

so

$$k_{11}(\rho)^2 = z_\rho^2 + 132 > \frac{577}{4} > 144, \quad k_{11}(\rho) > 12.$$

The upper wall $\rho < 1$ is essential: z_ρ is undefined at $\rho = 1$, and the unqualified assertion for all $\rho \geq \rho_c$ would be false, for example at $\rho = 2$.

Define the unique subtraction owners

$$\mathcal{C}_{21} = \mathcal{D}_{20} \cap \{K \leq 200\}, \quad \mathcal{T}_{21} = \mathcal{D}_{20} \cap \{K > 200\}.$$

Every point of $\mathcal{C}_{21}$ lies in the compact rectangle, and every point of $\mathcal{T}_{21}$ lies in the aggregate theorem. They are disjoint and exhaust $\mathcal{D}_{20}$, with $K = 200$ assigned only to $\mathcal{C}_{21}$. This remains true whether $U(\rho) < 200$, $U(\rho) = 200$, or $U(\rho) > 200$. Therefore

$$\boxed{\mathcal{D}_{21} := \mathcal{D}_{20} \setminus (\mathcal{C}_{21} \cup \mathcal{T}_{21}) = \emptyset.} \tag{4.57}$$

The four inherited faces retain their owners: $\rho = \rho_c$ is included subject to the two strict frequency inequalities, while $\rho = 39/50$, $K = k_{11}$, and $K = U$ are outside $\mathcal{D}_{20}$. The exact classifier printed in Supplement S1 enumerates all $9 \cdot 7 = 63$ listed ratio/frequency state pairs and five separate interface assertions, and verifies the same disjoint-union truth table. $\square$

Proposition 4.13 (Compact-middle Pólya theorem). *For every*

$$\rho_* < \rho < \frac{7}{8}, \quad K \geq 0,$$

the Dirichlet counting function of the concentric spherical shell satisfies

$$N_D(A_{\rho,1}, K^2) \leq W(\rho, K).$$

Proof. Proposition 4.1 proves the inequality outside the exact residual $\mathcal{D}_{16}$ in (4.11). Within that residual, Lemmas 4.2, 4.3, 4.4, 4.5, 4.6, 4.7, and 4.8 remove precisely the regions subtracted in the definition of $\mathcal{D}_{20}$. Hence any point not yet owned belongs to $\mathcal{D}_{20}$ in (4.38). The compact certificate proves the inequality on $\mathcal{C}_{21}$, and the aggregate certificate proves it on $\mathcal{T}_{21}$. These two sets are disjoint and exhaust $\mathcal{D}_{20}$, as the exact identity (4.57) shows. Thus no residual point remains. The cases on every threshold face have the explicit owners specified above; in particular, equality at a frequency wall is not lost in taking complements. Finally, $K = 0$ is immediate from $N_D(A_{\rho,1}, 0) = W(\rho, 0) = 0$. This proves the assertion for the whole stated parameter range. $\square$

4.8 Certificate protocol, reproducibility data, and coverage checklist

The finite computations above are interval proofs, not floating-point experiments. The proof-decision rules are:

1. all parameter endpoints, subdivisions, active-channel cutoffs, and set relations use exact integers or rationals;
2. all transcendental values use outward Arb balls at not less than 192-bit midpoint precision;
3. a sign is accepted only by a certain comparison of the complete ball with zero; a displayed midpoint or decimal is never a predicate;
4. strict integer walls use $[x]_< \leq M - 1$ after the certified comparison $x \leq M$, never $[x]$ at equality;
5. exact interfaces $\nu = \rho K$, $\nu = K$, $\rho = 1/2$, half-integer μ , and $K = 200$ have explicit owners;
6. every accepted run is repeated at higher precision, and each complete deterministic certificate is hashed canonically;
7. the complete replay sources are printed verbatim in Supplements S1 and S2. They read no input files, use only exact rational data and outward Arb balls, and exit nonzero on the first failed proof predicate.

The final exact replay reports are:

exact-rational ledger	611 checks; canonical registry; PASS
compact, 256 bits	10580 leaves; 2153076 floor comparisons; PASS
compact, 384 bits	10580 leaves; 2153076 floor comparisons; PASS
aggregate base, 192 bits	1286 boxes; 14146 signs; PASS
aggregate base, 384 bits	1286 boxes; 14146 signs; PASS
exact set logic	68 state/interface checks; disjoint union; PASS.

The exact-rational registry contains (488) substantive checks, (65) bookkeeping identities, and (58) supplemental checks. Its canonical registry SHA-256 is

afd7e054f52aa9a3992fc8ef16d3e755
1586c76bfb589053714f5f55a36792f1.

The SHA-256 digests of the three source listings are

exact-rational source	41691116e0e3e5453b9b209acb93ff78
	4724c3a95b83c94bbdf7f6760616e716
compact source	bc8c6771a03bde0d0560c41bb254a9db
	c128edb53a00555303132a3c1a25a9e5
aggregate source	e71dde74a259f6c55068c3a77e129898
	c22441fe708e3a20f21a65f66821c46e

Coverage checklist. The fragment contains: the exact primitive mask and both active upper switches; the global 295^2 owner; the fixed-channel shell-to-ball and angular comparisons; all first-, second-, and third-zero bounds used; the first, next, and two-sided staircases; every printed multiplicity inventory; the lower shifted-tail wedge and exceptional cap-395 cell; the high staircase through k_{11} , including the corrected cap-74 cell; both optical screens; the compact proxy implication; the aggregate discrete reserve, continuous reserve, derivatives, curvature propagation, and two integrations; the corrected $\rho < 1$ scope of $k_{11} > 12$; and exact ownership of $\rho_c, 39/50, k_{11}, U, 12, 200$.

Deterministically regenerated certificate data. The 10580 individual compact leaf records (four rational endpoints, depth, and integer proxy cap), the 1286 individual aggregate outward balls and their per-predicate enclosures, and the 2153076 individual Arb-versus-integer channel comparisons are too large to print as raw arrays. They are deterministic outputs regenerated by the complete sources printed in Supplement S2; they are not additional mathematical assumptions. The recorded counts, source digests, certificate digests, and replay results make the regeneration byte-identifiable. Independent verification should preferably use a separately written implementation or a second interval library. The digests identify runs; they are run metadata, not logical premises.

5 Proof of the spectral theorem

We now assemble the regional results without importing any new estimate. First, elementary bounds give

$$3\sqrt{3} > 5 > \pi,$$

and hence $\omega_0 > 0$. Also $\sqrt{3} < 2$ and $\pi > 3$ give $\omega_0 < 1/6$. Since $C_* > 0$,

$$0 < \rho_* < \omega_0 < \frac{1}{6} < \frac{7}{8}. \quad (5.1)$$

Therefore

$$(0, 1) = (0, \rho_*] \dot{\cup} (\rho_*, 7/8) \dot{\cup} [7/8, 1). \quad (5.2)$$

Fix $0 < \rho < 1$. At $K = 0$, positivity in Proposition 2.1 gives

$$N_D(A_{\rho,1}, 0) = 0 = W(\rho, 0). \quad (5.3)$$

Let $K > 0$. If $0 < \rho \leq \rho_*$, apply Theorem 2.9. If $\rho_* < \rho < 7/8$, apply Proposition 4.13. If $7/8 \leq \rho < 1$, apply Lemma 3.8. The three cases in (5.2) are disjoint and exhaustive, so

$$N_D(A_{\rho,1}, K^2) \leq \frac{2}{9\pi}(1 - \rho^3)K^3 \quad (K \geq 0). \quad (5.4)$$

Now

$$\omega_3 = \frac{4\pi}{3}, \quad L_3 = \frac{\omega_3}{(2\pi)^3} = \frac{1}{6\pi^2}, \quad |A_{\rho,1}| = \frac{4\pi}{3}(1 - \rho^3),$$

so

$$L_3|A_{\rho,1}| = \frac{2}{9\pi}(1 - \rho^3). \quad (5.5)$$

Taking $K = \sqrt{\Lambda}$ in (5.4) proves the unit-shell form of Theorem 1.1 for every $\Lambda \geq 0$.

For arbitrary $0 < r < R$, put $\rho = r/R$. The unitary dilation

$$(D_R u)(x) = R^{-3/2}u(x/R)$$

maps $L^2(A_{\rho,1})$ onto $L^2(A_{r,R})$, and its Dirichlet form scales by R^{-2} . Hence

$$\lambda_j^D(A_{r,R}) = R^{-2}\lambda_j^D(A_{\rho,1}),$$

and the strict threshold transforms as

$$N_D(A_{r,R}, \Lambda) = N_D(A_{\rho,1}, R^2\Lambda). \quad (5.6)$$

Applying the unit-shell result at $R^2\Lambda$,

$$\begin{aligned} N_D(A_{r,R}, \Lambda) &\leq \frac{2}{9\pi}(1 - \rho^3)(R^2\Lambda)^{3/2} \\ &= \frac{2}{9\pi}(R^3 - r^3)\Lambda^{3/2} \\ &= L_3|A_{r,R}|\Lambda^{3/2}. \end{aligned}$$

This proves Theorem 1.1 for every $R > 0$, including $R < 1$.

6 Proof that spherical shells do not tile

We prove Theorem 1.2 independently of the spectral argument. Set $\alpha = r/R \in (0, 1)$ and scale by $1/R$. It suffices to consider

$$T = A_{\alpha,1} = \{x : \alpha < |x| < 1\}.$$

Every rigid motion has the form $x \mapsto Qx + a$, with Q orthogonal. Because T is radial, $Q(T) = T$, so every congruent copy is a translate

$$T_a = a + T = \{x : \alpha < |x - a| < 1\}.$$

Assume for contradiction that $\{T_a : a \in \mathfrak{C}\}$ has pairwise-disjoint interiors and covers almost everywhere. Duplicate centers are impossible because they give the same nonempty open tile.

Fix a unit vector e and put

$$c = \frac{1 + \alpha}{2}e, \quad \delta = \frac{1 - \alpha}{2}.$$

Write $B(x, s) = \{y \in \mathbb{R}^3 : |y - x| < s\}$ for the open Euclidean ball. Then $B(c, \delta) \subset T$. Therefore T_a contains $B(a + c, \delta)$, and disjointness implies

$$|a - b| \geq 1 - \alpha \quad (a \neq b). \quad (6.1)$$

Thus the centers are locally finite: a bounded set cannot contain infinitely many points with fixed positive separation. They are also countable, since

$$\mathfrak{C} = \bigcup_{m \geq 1} (\mathfrak{C} \cap \overline{B}(0, m))$$

is a countable union of finite sets.

Each tile boundary is a union of two spheres and is three-dimensional null. The countable union

$$Z = \bigcup_{a \in \mathcal{C}} \partial T_a$$

is therefore null. Exact open-shell coverage is already a special case of almost-everywhere coverage. If closed shells cover exactly or almost everywhere, removing Z shows that their open interiors still cover almost everywhere. Hence it is enough to assume

$$\left| \mathbb{R}^3 \setminus \bigcup_{a \in \mathcal{C}} T_a \right| = 0. \quad (6.2)$$

Choose one tile T_{a_0} and its outer sphere

$$\Sigma = \{x : |x - a_0| = 1\}.$$

Fix $p \in \Sigma$ and put $u = p - a_0$. For $n \geq 1$, let

$$U_n = B\left(p + \frac{2}{n}u, \frac{1}{n}\right).$$

Every U_n lies outside T_{a_0} , is open, and has positive volume. By (6.2), choose $x_n \in U_n \cap T_{a_n}$ with $a_n \neq a_0$. Since $x_n \rightarrow p$ and $|x_n - a_n| < 1$, the centers a_n eventually lie in a fixed bounded set. Local finiteness lets us pass to a subsequence with one fixed center $a \neq a_0$. Thus $p \in \overline{T_a}$.

The point p is not in T_a . Otherwise some $B(p, \varepsilon)$ lies in T_a . Choose

$$0 < t < \min\{\varepsilon, 1 - \alpha\}.$$

Then $p - tu \in T_{a_0} \cap T_a$, contradicting disjoint interiors. Therefore every $p \in \Sigma$ belongs to the boundary of a neighboring tile.

If $p \in \Sigma \cap \partial T_a$, then $|p - a| \in \{\alpha, 1\}$, hence $|a - a_0| \leq 2$. Only finitely many centers satisfy this bound. Consequently Σ is covered by finitely many neighboring inner or outer boundary spheres.

Translate so that $a_0 = 0$. The intersection of Σ with a neighboring sphere $\{x : |x - a| = s\}$, $s \in \{\alpha, 1\}$, satisfies

$$|x|^2 = 1, \quad |x - a|^2 = s^2,$$

and hence

$$2a \cdot x = |a|^2 + 1 - s^2. \quad (6.3)$$

Because $a \neq 0$, (6.3) is a genuine plane. Its intersection with Σ is empty, a tangent point, or a circle, and therefore has zero surface measure on Σ . A finite union of such sets cannot cover Σ , whose area is 4π .

There is no coincident-sphere exception. A neighboring outer sphere equals Σ only when it has the same center, which is the original tile or a forbidden duplicate. A neighboring inner sphere cannot equal Σ because $\alpha < 1$. We have reached a contradiction. Scaling back proves Theorem 1.2.

A External inputs and verification boundary

The proof above is self-contained with respect to every shell-specific spectral reduction, weighted count, endpoint estimate, regional subtraction, fixed-channel variational comparison, seam assignment, and certificate predicate. It uses the following conventional external results, each only in the scope stated where it enters the proof:

1. standard spherical-harmonic decomposition, regular Sturm–Liouville theory, and the classical Bessel identities collected in [5];
2. the uniform one-variable Bessel-phase enclosure of [2], together with the explicitly quoted floor-sum and turning-point inequalities of [1, 3];
3. Lorch’s strict lower bound for the first positive zero of J_ν , in precisely the half-integer specializations derived in the text [4]; and
4. outward midpoint-radius interval arithmetic as implemented by Arb [6]. The mathematical proof predicates, rational partitions, precision gates, coverage rules, strict-wall handling, and deterministic regeneration algorithm are all stated in the main paper, while the complete executable sources are printed in the accompanying certification supplement. Decimal displays are never proof predicates.

The computer-assisted lemmas are conditional only on the correctness of the small trusted arithmetic kernel that implements exact rational arithmetic and outward Arb elementary functions. The leaf and box arrays are deterministic outputs of the supplementary algorithms, not additional mathematical assumptions. Independent replay should nevertheless use a separately built implementation or a second interval library before external publication.

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